MODULE 1 : Intercorporate Investments

Student's notes

Student's notes

LEARNING OUTCOMES

1.a. Describe the classification, measurement, and disclosure under International Financial Reporting Standards (IFRS) for 1) investments in financial assets, 2) investments in associates, 3) joint ventures, 4) business combinations, and 5) special purpose and variable interest entities

1.b. Compare and contrast IFRS and US GAAP in their classification, measurement, and disclosure of investments in financial assets, investments in associates, joint ventures, business combinations, and special purpose and variable interest entities

1.c. Analyze how different methods used to account for intercorporate investments affect financial statements and ratios

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and comparison to US GAAP

Classification for intercorporate investments

Features	Financial assets	Associate	Joint Venture	Consolidation
Ownership	<20%	20%-50%	Shared control	>50%
Influence	No significant influence	Significant influence	Joint control	Control
Accounting	Cost and market value	Equity method	Equity method (both GAAP & IFRS) Pro-rata consolidation (rare)	Consolidation
Income to parent	Dividends, interests and capital gains	Earnings combined to extent of ownership, irrespective of actual payments	Same as Equity Method	100% revenue and expenses are added. Intercompany transactions are eliminated.

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and….

Financial investments	Amortized cost	
	Fair value	FVOCI
		FVPL
Investments in associates	Equity method	
Joint ventures	Equity method & Proportionate consolidation method (rare)	
Business combinations	Acquisition method	

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

General recognition, measurement, disclosure and presentation for financial assets (IFRS 9)

1. Recognition	**Recognition** A financial instrument is recognized in the financial statements when the entity becomes a party to the financial instrument contract.
	Derecognition An entity removes a financial asset from its statement of financial position when: • Its contractual rights to the asset's cash flows expire; • It has transferred the asset and substantially all the risks and rewards of ownership.
2. Measurement	**At the date of acquisition** All financial assets are measured at fair value when initially acquired.
	In subsequent periods Financial assets are measured at either fair value or amortized cost.

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

General recognition, measurement, disclosure and presentation for financial assets (IFRS 9)

3. Presentation	Financial instruments are classified into • financial assets, • financial liabilities and; • equity instruments.
4. Disclosures	The significance of financial instruments for the entity's financial position and performance
	The nature and extent of risks arising from financial instruments to which the entity is exposed during the period

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Investment in financial assets		
Accounting for financial assets	Amortized cost **(1)**	
	Fair value	Fair value through profit or loss (FVPL) **(2)**
		Fair value through other comprehensive income (FVOCI) **(3)**

At the date of acquisition, all financial assets are measured at fair value when initially acquired.
In subsequent periods, financial assets are measured at either fair value or amortized cost.

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

(1) Amortized Cost (for Debt Securities Only)

Criteria for amortized cost accounting:

1. Business model test: securities held to collect contractual cash flows.
2. Cash flow characteristic test: the contractual cash flows are principal or interest only

- Amortized cost is the original cost of the debt security + **discount**, or - **premium** that has been amortized to date.
- **Interest income** (coupon cash flow adjusted for amortization of premium or discount) is recognized in the income statement, but subsequent **changes in fair value** are ignored.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

(2) Fair value through Profit or Loss

- **Debt securities held for trading** may be classified as fair value through profit or loss (FVPL), or if accounting for those securities at amortized cost results in an accounting mismatch.
- **Equity securities that are held for trading** must be classified as FVPL. Other equity securities may be classified as either FVPL, or FVOCI. Once classified, the choice is *irrevocable*.
- **Derivatives that are not used for hedging** are always carried at FVPL. If an asset has an embedded derivative (e.g., convertible bonds), the asset as a whole is valued at FVPL.
- ➢ FVPL securities are reported on the balance sheet at fair value →The changes in fair value, both realized and unrealized & dividend or interest income are recognized in the income statement.

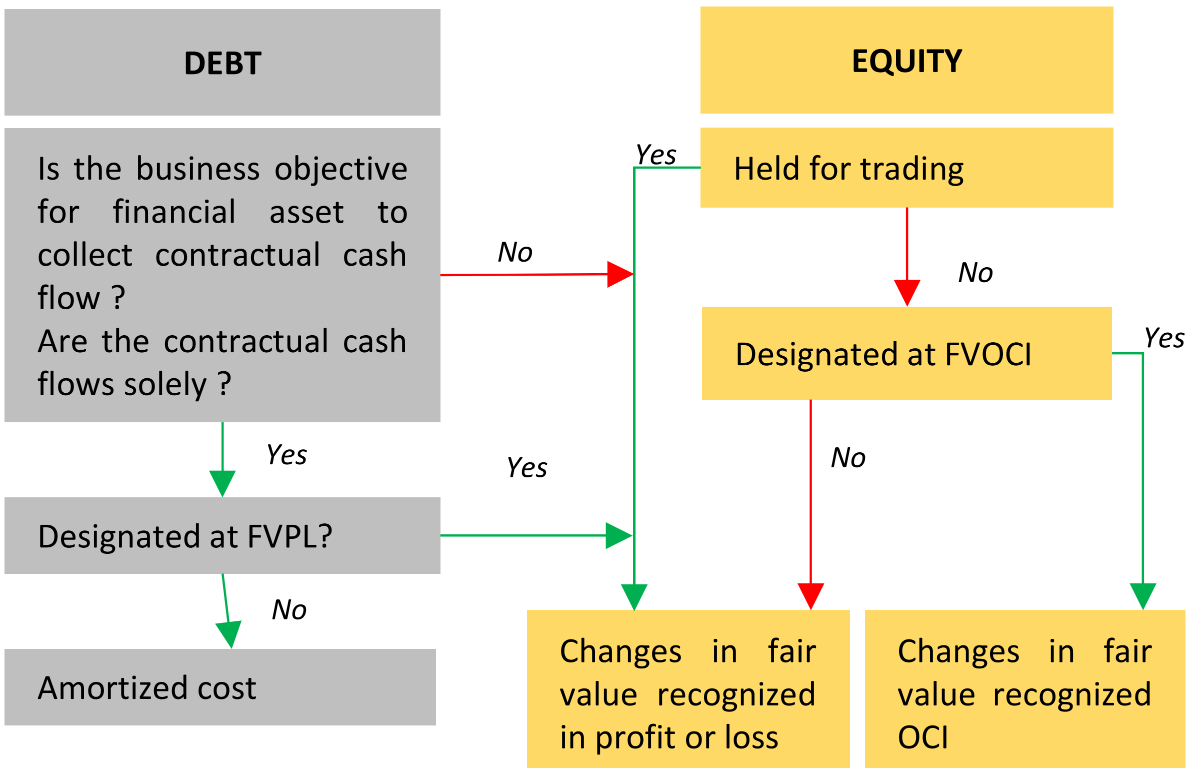
(3) Fair Value Through Other Comprehensive income

- Carried on the balance sheet at fair value → unrealized gain or loss is reported in OCI >< Realized gain or loss, dividends, and interest income are reported in the income statement.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Classification of financial assets



Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Detailed accounting treatment for investment in financial assets (IFRS 9)

	Amortized Cost	Fair Value Through Profit or Loss	Fair Value Through OCI
Balance sheet	Amortized cost	Fair value	Fair value, with G/L recognized in equity
Income statement	Interest (including amortization) Realized G/L	Interest Dividends G/ L	Interest Dividends

Example: At the beginning of the year, Midland Corporation purchased a 9% bond with a face value of $100,000 for $96,209 to yield 10%. The coupon payments are made annually at year-end. Suppose that the fair value of the bond at the end of the year is $98,500.

Determine the impact on Midland's balance sheet and income statement if the bond investment is classified as

1) amortized cost

2) fair value through profit or loss, and

3) fair value through OCI

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Example of accounting treatment for investment in financial assets (IFRS 9)

	Amortized cost	Fair value	
		FVPL	FVOCI
Balance sheet (opening)	Original cost = FV_0 = 96,209	Fair value = FV_0 = 96,209	Fair value = FV_0 = 96,209
Income statement	Interest income = 9,621 = 96,209*10%	Interest income = 9,621 = 96,209*10%	Interest income = 9,621 = 96,209*10%
Balance sheet	Amortized cost = 96,209 + 9,621 – 100,000*9% = 96,830	Fair value = 98,500	Fair value = 98,500
Realized/ unrealized gain	N/A	Unrealized gain = 98,500 – 96,209 – 621 = 1,670 in P&L	Unrealized gain = 98,500 – 96,209 – 621 = 1,670 in OCI

Question:
Now let's imagine that the bonds are called on the first day of the next year for $101,000. **Calculate the gain or loss recognition for each classification.**

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Example of accounting treatment for investment in financial assets (IFRS 9)

	Amortized cost	Fair value	
		FVPL	FVOCI
Balance sheet value	Amortized cost = 96,830	Fair value = 98,500	Fair value = 98,500
Sale price	101,000	101,000	101,000
Realized gain	4,170 = 101,000 – 96,830 in IS	2,500 = 101,000 – 98,500 in IS	4,170 = 101,000 – 96,830 in IS
Unrealized gain	N/A	N/A	Unrealized gain 1,670 is removed from equity

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

IFRS 9 New Standards

• **Reclassification of equity securities not permitted** - initial designation is **irrevocable** (FVPL or FVOCI). • Reclassification of debt securities is permitted **only if** the business model has changed.	• Loss model for loan impairment was replaced by the **expected credit loss model.** • Companies are required not only to evaluate current and historical information about loan but to also use forward-looking information.

Financial asset impairment

US GAAP: a security is considered impaired if its decline in value is determined to be other than temporary.	**IFRS:** impairment of a debt or equity security is indicated if at least one loss event has occurred, and its effect on the security's future cash flows can be estimated reliably.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Equity method

The value of the investment on the balance sheet

(Carrying value is a Non- current asset)

=

Initial investment (cost to acquire)

+

The investor's proportionate earning/loss in the investee (also recognized in the investor's income statement)

−

Dividends

Note: If the investee's losses reduce the investment value < 0, the investor usually **discontinues** use of the equity method. The equity method is **resumed** once the proportionate share of the investee's earnings > the share of losses that were not recognized during the suspension period.

Fair Value Option	**U.S. GAAP** allows equity method investments to be recorded at fair value
	Under IFRS: only available to venture capital firms, mutual funds, and similar entities.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Question

Suppose that we are given the following:

- December 31, 20X5, Company P (the investor) invests $1,000 in return for 30% of the common shares of Company S (the investee).
- During 20X6, Company S earns $400 and pays dividends of $100.

Calculate the effects of the investment on Company P's balance sheet, reported income, and cash flow for 20X6

Answer

Value of the investment (on B/S) = **Initial investment** + **Proportionate P/L** (on IS) **- Dividends**

Company P will:

- Recognize $120 ($400 × 30%) in the income statement from its proportionate share of the net income of Company S.
- Increase its investment account on the balance sheet by $120 to $1,120
- Increase cash by $30 (dividends) and reduce the investment account by the same amount

→ The carrying value of the investment at year-end = $1000 (initial investment) + $120 (proportionate share of Company S' net income) - $30 (dividends received) = $1090

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Excess of purchases price over book value acquired

At the acquisition date	In subsequent periods
The excess is allocated to identifiable assets and liabilities **based on fair value.** Any remainder is considered goodwill	Excess amount recognized as expense consistent with the investee's recognition of expense

Amortization of Excess Purchase Price:

- Amounts allocated to inventory are expensed.
- Amounts allocated to depreciable or amortizable assets are capitalized and subsequently expensed (depreciated or amortized) over an appropriate period of time.
- Amounts allocated to land and other assets or liabilities that are not amortized continue to be reported at fair value as of the date of investment.

Note:

- The purchase price allocation to the investee's assets and liabilities is included in the investor's balance sheet, not the investee's.
- The additional expense that results from the assigned amounts is not recognized in the investee's income statement.
- The investor must adjust its balance sheet investment account and the proportionate share of the income reported from the investee for this additional expense.

Student's notes

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Question

On January I, 2010, Prime Manufacturers acquired a 25% equity interest in Alton Corp. for $700,000. Plant and equipment are depreciated on a straight-line basis to zero over a term of 10 years. Prime uses the equity method to account for its investment in Alton. Alton reports net income of $250,000 for 2010 and pays dividends of $100,000. Calculate the following:

1. Goodwill included in the purchase price.
2. The amount of equity income to be reported on Prime's income statement for 2010.
3. The value of its investment in Alton recognized by Prime on its balance sheet for 2010

	Book value ($)	Fair value ($)
Current assets	220,000	220,000
Plant and equipment	1,300,000	1,500,000
Land	900,000	1,050,000
Total assets	2,420,000	2,770,000
Liabilities	750,000	750,000
Net assets	1,670,000	2,020,000

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Answer

1. Proportionate share in book value of Alton's net assets = 25% x 1,670,000 = 417,500 < 700,000: purchase price → allocation to the investee's identifiable assets and liabilities (plant and equipment, land) **based on fair value.**

Allocation to plant and equipment: = 25% (fair value – book value) = 50,000
Allocation to land = 25% (land fair value – land book value) = 37,500
Any remainder is goodwill → Goodwill = 700,000-417,500-50,000-37500= $195,000

2. Proportionate share in Alton's 2010 earnings = 25% X $250,000 = $62,500

Allocation principles: The amount of equity income to be reported on Prime's income statement for 2010.

⇒ Amortization allocation (Expense) to plant and equipment = 50,000 (total allocation to plant and equipment) / 10 years (useful life) = 5,000

⇒ Equity income for investor (Prime manufactures) = 62,500 – 5000 = 57,500

3. The value of its investment in Alton recognized by Prime on its balance sheet for 2010

= purchase price (initial investment)+ equity income – proportionate dividends (according to equity method) = 700,000+57,500-(25%x100,000) = 732,500

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Impairment

US GAAP: when fair value < carrying value and considered as permanent
IFRS: when 1 or more loss event happens
Both IFRS and US GAAP: written-up is not allowed

Fair value option for equity method

US GAAP: allowed
IFRS: only allowed for venture capital firms, mutual funds, and similar entities
The decision to use fair value option is **irrevocable**

Fair value option accounting

- Unrealized gains/losses arising from changes in fair value as well as interest and dividends received are included in the investor's income.
- The investment account on the investor's balance sheet does not reflect the investor's proportionate share in the investee's earnings. dividends, or other distributions.
- The excess of cost over the fair value of the investee's identifiable net assets is not amortized.
- Goodwill is not created.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Investment in associates – Transactions with the investee

Investee → Upstream transaction → Investor

Features

Investee: recognize all of the profit in its income statement.

Investor: eliminate its proportionate share of the **unconfirmed** profit from the equity income of the investee.

Investee ← Downstream transaction ← Investor

Features

Investor: Recognize all of the profit in its income statement and eliminate the proportionate share of the profit that is unconfirmed.

Note: Because of its ownership interest, the investor may be able to influence transactions with the investee. Thus, profit from these transactions must be deferred until the profit is confirmed through use or sale to a third party.

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Example

Prime Manufactures and Alton: Transaction

- 1. $12,000 of profit from an upstream sale from Alton to Prime during 2010 was still in Prime's inventory at the end of 2010 as the goods had not yet been sold to an outside party.
- 2. Prime made downstream sales of $ I00,000 worth of goods to Alton for $160,000. During 2010, Alton sold goods worth $140,000 to outside parties, while the remaining $20,000 worth of goods was sold in 2011

Calculate the amount of equity income reported on Prime's income statement for 2010 and the value of the investment in Alton on Prime's 2010 balance sheet after incorporating the effects of the above transactions.

Student's notes

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Answer

Upstream sale:

- Unrealized profit= $12,000
- Prime's proportionate share in the unrealized profit= 25% x 12,000 = $3,000

Downstream sale:

- Prime's profit on sales to Alton = 160,000 - 100,000 = $60,000

Alton sells 87.5% (=140,000/ 160,000 x 100) of goods purchased from Prime. During

- 2010, while 12.5% remains unsold → total unrealized profit= 12.5% x 60,000 = $7,500

→ Prime's **proportionate share** of unrealized profit = 25% x 7,500 = $1,875 .

Therefore:

- Revised equity income for 2010 = 57 ,500 - 3,000 - 1,875 = $52,625 .
- Revised carrying value of investment for 20 I 0 = 732,500 - 3,000 - 1,875 = $727,625
- In 2011, when unconfirmed sales for 2010 (worth $20,000) are actually confirmed related profits that were not realized during 2010 (worth $1,875) are realized. These profits contribute to equity income for 2011.

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Business combinations

Merger

Company A + Company B = **Company A**

Acquisition

Company A + Company B = **(Company A + Company B)**

Note:
The distinctive feature of an acquisition is the legal continuity of the entities. Each entity continues operations but is connected through a parent–subsidiary relationship

Consolidation

Company A + Company B = **Company C**

Note:
The distinctive feature of a consolidation is that a new legal entity is formed and none of the predecessor entities remain in existence. A new entity is created to take over the net assets of Company A and Company B.

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method

The parent's financial statements (book value)

+

Its subsidiaries financial statements (fair value)

Consolidated financial statements

Items that need to be combined:

- Liabilities
- Assets
- Revenues
- Expenses

Note:

Transactions between the parent and subsidiary **(intercompany transactions)** are eliminated to avoid double counting and premature income recognition.

If a parent **owns less than a 100%** equity interest in the company, it must also create a **non controlling interest** account on the consolidated balance sheet and income statement

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

	IFRS	US GAAP
Identifiable Assets and Liabilities	The acquirer: - measure the identifiable tangible and intangible assets and liabilities of the acquiree **at fair value** as of the date of the acquisition - recognize any assets and liabilities that the acquiree had not previously recognized (brand names, patent,….)	
Contingent Liabilities	The acquirer: include contingent liabilities if their fair values can be reliably measured	The acquirer: includes only those contingent liabilities that are probable and can be reasonably estimated.
Indemnification Assets	The acquirer must recognize an indemnification asset if the acquiree contractually indemnifies the acquirer for the outcome of a contingency or an uncertainty related to all or part of a specific asset or liability of the acquiree.	
Financial Assets and Liabilities	Identifiable assets and liabilities acquired are classified in accordance with IFRS (or US GAAP) standards. The acquirer can reclassify financial assets and liabilities.	
Price is Less than Fair Value	The difference between the fair value of the acquired net assets and the purchase price to be recognized immediately as a gain in profit or loss.	

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (when the parent company owns 100%)

	Pyramid Inc.	Sam Corp	
	Book value	*Book value*	*Fair value*
Cash and receivables	12,500	450	450
Inventory	11,000	1,500	3,500
Net PP&E	25,500	2,800	4,000
Total assets	**49,000**	**4,750**	**7,950**
Current payables	7,500	800	800
Long term debt	15,000	2,500	2,000
Total liabilities	**22,500**	**3,300**	**2,800**
Net assets	**26,500**	**1,450**	**5,150**
Capital stock ($1 par)	5,500	300	
Additional paid-in capital	7,000	400	
Retained earnings	14,000	750	
Total equity	**26,500**	**1,450**	

Student's notes

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and...

Acquisition method example (when the parent company owns 100%)

Question

Pyramid Inc. acquired a **100%** equity interest in Sam Corp. by issuing 1 million shares of common stock. The par value of each share was $1, while the market price of each share at the time of the transaction was $10. Based on the **acquisition method**, calculate the amounts presented on the post-combination balance sheet. Assume that Sam has no identifiable intangible assets.

Answer

Step 1: Calculated excess purchase price

Purchase price = money raised from shared issue = 10,000,000

→ Excess purchase price = purchase price (10,000,000) – book value of the acquire (1,450,000) = 8,550,000

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (when the parent company owns 100%)

Answer

Step 2: Allocate excess purchase price to identifiable the acquiree's identifiable **net asset**, liabilities (similar to equity method except that the acquirer allocate to the acquiree's net assets, not **proportionate allocation**)

Allocation to inventory = 2,000,000 (increase in asset)

Allocation to PPE = 1,200,000 (increase in asset)

Allocation long term debt = 500,000 (decrease in debt)

Goodwill = 8,550,000 – 500,000 – 200,000 – 1,200,000 = 4,850,000

Step 3: Calculate equity of the consolidated company after issuing stock

Capital stock = Parent's capital stock (5,500,000) + Par value of shares issued (1,000,000) = $6,500,000

Additional paid-in capital = Parent's additional paid-in + Market value of shares issued - Par value of shares issued = 7,000,000 + [(1,000,000 x 10)-(1,000,000 x 1)] = $16,000,000

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Consolidated balance sheet	$'000
Cash and receivables = (12,500 + 450)	12,950
Inventory = (11,000 + 2,000 + 1,500)	14,500
Net PP&E = (25,000 + 2,800 + 1,200)	29,500
Goodwill	4,850
Total assets	**61,800**
Current payables (7,500 + 800)	8,300
Long term debt (15,000 + 2,500 - 500)	17,000
Total liabilities	**25,300**
Capital stock ($1 par) (5,500 + 1,000)	6,500
Additional paid-in capital (7,000 + 9,000)	16,000
Retained earnings	14,000
Total equity	**36,500**

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Non controlling interest (appear when the parent owns < 100%)

The parent **equity**	Total net income (in case the parent owns 100%)
+	−
NCI on the **BS** of the consolidated company	**NCI** on the **IS** of the consolidated company
The consolidated company's **equity**	The consolidated company's **net income**

NCI calculation:

- NCI on the BS of the parent company = Minority shareholders proportionate share x Acquiree's equity
- NCI on the IS of the parent company = Minority shareholders proportionate share x Acquiree's net income

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (When the parent company owns < 100%)

Question

Suppose Company P acquires **80%** of the common stock of Company S on December 31, 2016, by paying **$120,000** cash to the shareholders of Company S. The two firms' preacquisition balance sheets as of December 31, 2016, and pre-acquisition pro forma income statements for the year ending December 31, 2017.

Calculate:

1. Company P's total assets after acquisition ?
2. Company P's pro forma consolidated net income ?
3. Company P's minority ownership interest on the balance sheet on December 31, 2017 ?

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (When the parent company owns < 100%)

Pre Acquisition Balance Sheets December 31,2016	Company P	Company S
Current assets	720,000	240,000
Other assets	480,000	120,000
Total assets	1,200,000	360,000
Current liabilities	600,000	210,000
Common stock	420,000	90,000
Retained earnings	180,000	60,000
Total liabilities & equity	1,200,000	360,000

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (When the parent company owns < 100%)

Unconsolidated Income Statements December 31,2017	Company P	Company S
Revenue	900,000	300,000
Expenses	600,000	240,000
Net income	300,000	60,000
Dividends		15,000

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[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Acquisition method example (When the parent company owns < 100%)

Answer

1. **Company P consolidated total asset (on December 31, 2016)** = Company P's total assets + Company S's total assets – Cash company P spent on the acquisition = $1,200,000 + $360,00 - $120,000 = $1,440,000

2. **Minority interest income** = Company S minority share (after being consolidated) x Company S net income = 20% x $60,000 = $12,000

Company P consolidated net income *on* December 31, 2017 = Company P + Company S - minority interest = $300,000 + $60,000 - $12,000 = $348,000

3. **Minority interest (on the BS) at the beginning of 2017** = Company S minority share x company S equity

Minority interest at the end of 2017 = Minority interest at the beginning of 2017 + (company S minority share x Company S net income in 2017) - (company S minority share x dividends paid) = $30,000 + (20% x 60,000) – (20% x 15,000) = $39,000

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Goodwill in consolidations

Full good will = **Fair value of the subsidiary** = purchase price/proportionate ownership − Fair value of net assets

Partial good will
= proportionate ownership x full goodwill

	IFRS	US.GAPP
Full good will	Allowed	Required
Partial good will	Allowed	Not allowed

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Impairment of goodwill

IFRS

Impairment loss = Carrying value − Recoverable amount of the cash-generating unit

Carrying value: including the goodwill that has been allocated to that unit.

Recoverable amount of the cash-generating unit: Max (net selling price, Value in use).

Net selling price: Asset fair value – Cost to sell

Value in use: The present value of the future cash flows

Student's notes

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Impairment of goodwill

US GAAP

Step 1: identification of impairment

Carrying value > Fair value

Impairment loss exists

Step 2: measurement of the loss

Impairment loss

Carrying amount − Implied fair value of goodwill

Fair value of the reporting unit − Fair value of the reporting unit's assets and liabilities

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Impairment of goodwill example

Question

Last year, Parent Company acquired Sub Company for $1,000,000. On the date of acquisition, the fair value of Sub's net assets was $800,000. Thus, Parent reported acquisition goodwill of $200,000 ($1,000,000 purchase price – $800,000 fair value of Sub's net assets).

At the end of this year, the fair value of Sub is $950,000, and the fair value of Sub's net assets is $775,000. Assuming the carrying value of Sub is $980,000, determine if an impairment exists and calculate the loss (if applicable) under U.S. GAAP and under IFRS.

Answer

U.S. GAAP (two-step approach):

1. Carrying value of Sub > Fair value ($980,000 > $950,000) => **an impairment exists.**
2. Measure the impairment loss: At the impairment measurement date,

Implied value of the goodwill = $950,000 fair value of Sub – $775,000 fair value of Sub's net assets) = $175,000 < $200,000 Goodwill carrying value → impairment loss of $25,000 is recognized → reducing goodwill to $175,000.

IFRS (one-step approach):

Goodwill impairment and loss under IFRS is 980,000 (carrying value) – 950,000 (fair value) = $30,000.

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Special purpose and variable interest entities

SPEs are often structured such that the sponsor company has control over the SPE's finances or operating activities while third parties have controlling interest in the SPE's equity.

They can be abused to obtain off-balance sheet financing to improve financial performance

The formation of the term variable interest entity "VIE" which is treated as a subsidiary for the sponsor company

A VIE is an entity that has one or both of the following characteristics:

1. At-risk equity that is insufficient to finance the entity's activities without additional financial support.
2. Equity investors that lack any one of the following:

- Decision making right
- The obligation to absorb expected losses
- The right to receive expected residual returns

Student's notes

Student's notes

[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

Joint venture	
Features	
Shared by 2 or more investors	Created in various legal, operating, and accounting forms

Example: Google's parent company and the pharma company Glaxo and Smith decided to enter into a joint venture agreement to produce bioelectric medicines the ratio of the ownership was 45%-55%. The joint venture lasted and was committed for 7 years with a capital of Euro 540 million.

Accounting methods used	**Equity method**
	The proportionate consolidation method (in rare circumstance): similar to a business acquisition, except the investor (often venturers) only reports the proportionate share of the assets, liabilities, revenues, and expenses of the joint venture

Student's notes

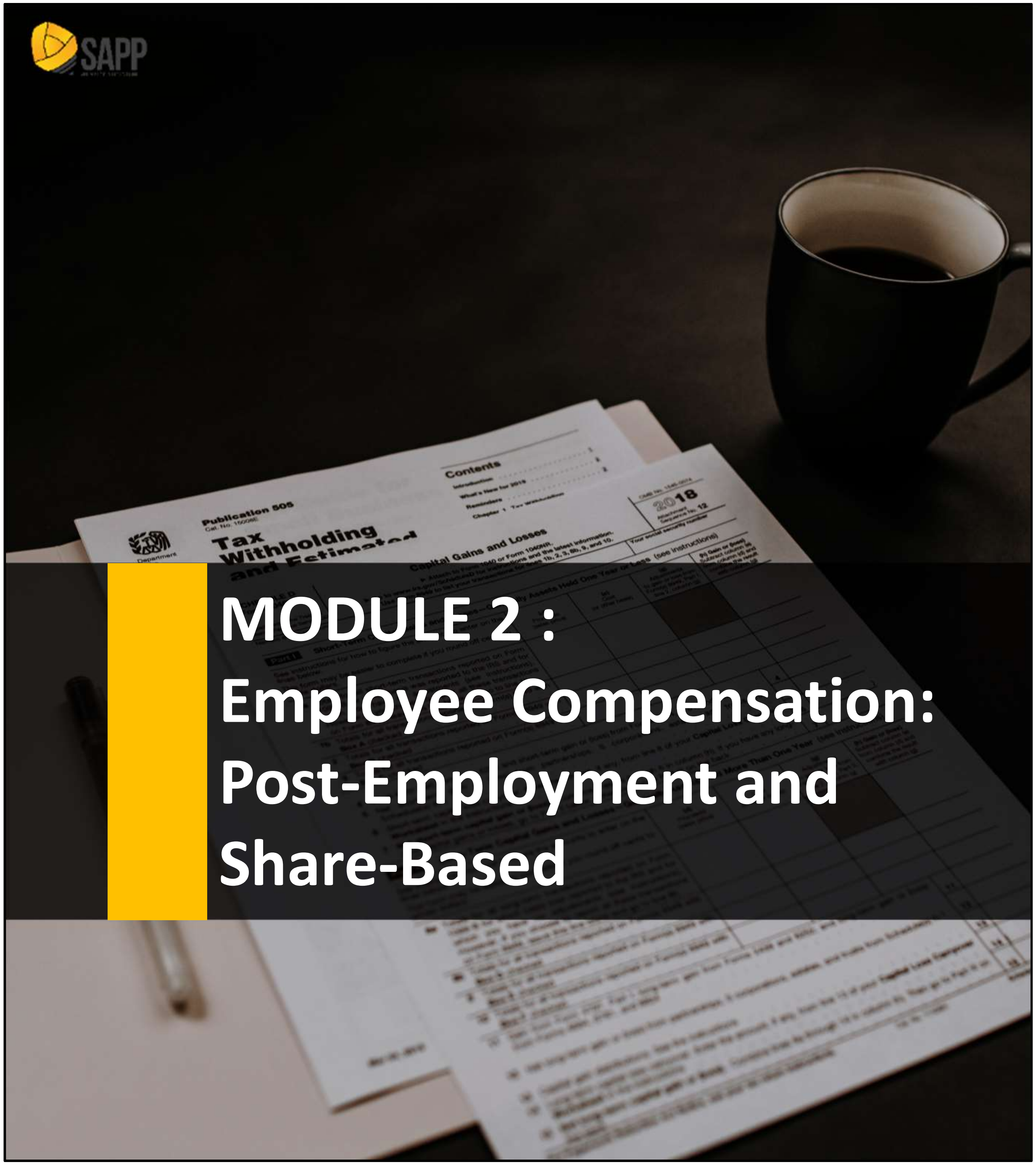
[LOS 1.a,b] Describe the classification, measurement and disclosure for intercorporate investments under IFRS and…

	IFRS	US GAAP
Contingent Assets and Liabilities	The cost of an acquisition is allocated to the fair value of assets, liabilities, and contingent liabilities. Subsequently, the contingent liability = max (the amount initially recognized, the best estimate of the amount required to settle)	Contractual contingent assets and liabilities are recognized and recorded at their fair values at the time of acquisition. Subsequently, contingent liability = max (the amount initially recognized, the best estimate of the amount of the loss), contingent asset = min (the acquisition date fair value, the best estimate of the future settlement amount)
Contingent Consideration	Initially measured at fair value. In subsequent periods, changes in the fair value of liabilities (and assets, in the case of US GAAP) are recognized in the consolidated income statement	
In-Process R&D	In-process research and development is recognized as a separate intangible asset and measure it at fair value	
Restructuring Costs	Recognized as an expense in the periods the restructuring costs are incurred.	

Student's notes

[LOS 1.c] Analyze how different methods used to account for intercorporate investments affect financial statements and ratios

	Equity method	Acquisition method
Leverage	Both liabilities and equity will be lower under the equity method. Impact on leverage will depend on which is affected more proportionately.	Both liabilities and equity will be higher under the equity method. Impact on leverage will depend on which is affected more proportionately.
Net profit margin	Higher – sales are lower, net income is the same	Lower - sales are higher and net income is the same
ROE	Higher – equity is lower, net income is the same	Lower – equity is higher, net income is the same
ROA	Higher – net income is the same and assets are lower	Lower, net income is the same, assets are higher



MODULE 2 : Employee Compensation: Post-Employment and Share-Based

Student's notes

Student's notes

LEARNING OUTCOMES

2.a. Contrast types of employee compensation

2.b. Explain how share-based compensation affects the financial statements

2.c. Explain how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

2.d. Explain how post-employment benefits affect the financial statements.

2.e. Explain financial modeling and valuation considerations for post-employment benefits

Student's notes

[LOS 2.a,b,c] Contrast types of employee compensation; Explain how share-based compensation affects the financial statements; Explain how to forecast share-based compensation expense and shares outstanding in a financial statement model and their use in valuation

Share-based compensation plans can take several forms, including **stock options** and **outright share grants**.

Advantage	Aligns employees' interests with those of shareholders → typically a form of deferred compensation
	Does not typically require a cash outlay.
Disadvantage	The employee may have limited influence on the company's market value, so share-based compensation may not necessarily provide the desired incentives.
	Increased ownership may make managers more risk-averse and conservative in their strategies.
	The option-like (asymmetrical) payoff may encourage management to take excessive risk.
	Leads to dilution of existing shareholder ownership.

Student's notes

[LOS 2.a,b,c] Contrast types of employee compensation; Explain how share-based compensation affects...

Stock grants	Stock appreciation right
Classification • Outright stock grant: shares are granted to the employee without any conditions. • Restricted stock grants: require the employee to return the shares to the company if certain conditions are not met • Performance share: grants are based on performance **Accounting for stock grants** Stock grants expense: based on the fair value of shares, which typically **equals their market value** on the grant date and allocated over the employee's service period.	**Features:** With stock appreciation rights (SARs) an employee's compensation is tied to increases in the company's stock price. but the employee does not own the company's stock. **Accounting for SARs:** Compensation expense for SARs is measured at fair value and allocated over the service life of the employee. (**Phantom stock**: Similar to stock appreciation rights except the payoff is based on the performance of hypothetical stock instead of the firm's actual shares)

Student's notes

[LOS 2.a,b,c] Contrast types of employee compensation; Explain how share-based compensation affects…

Stock options	Option pricing model
Accounting for stock options Stock options expense: based on the fair value of the options , which typically are estimated by using **a valuation model** and allocated in the income statement over the employee's service period.	The valuation method should (1) be consistent with fair value measurement (2) be based on established principles of financial economic theory (3) reflect all substantive characteristics of the award. Option-pricing models typically incorporate the following six inputs: • Exercise price. • Stock price at the grant date. • Expected term. • Expected volatility. • Expected dividends. • Risk-free rate.

Student's notes

[LOS 2.a,b,c] Contrast types of employee compensation; Explain how share-based compensation affects…

Stock options	Option pricing model
Accounting for stock options Stock options expense: based on the fair value of the options , which typically are estimated by using **a valuation model** and allocated in the income statement over the employee's service period.	The valuation method should (1) be consistent with fair value measurement (2) be based on established principles of financial economic theory (3) reflect all substantive characteristics of the award. Option-pricing models typically incorporate the following six inputs: • Exercise price. • Stock price at the grant date. • Expected term. • Expected volatility. • Expected dividends. • Risk-free rate.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Type of Benefit	Amount of Benefit	Obligation of Sponsoring Company	Sponsoring Company's Pre-funding
Defined-contribution pension plan	• Amount of future benefit is **not defined**. • Actual future benefit will depend on investment performance. • Investment **risk is borne by employee.**	Amount of contribution is **defined** each period with no additional future obligation	**Not applicable**
Defined-benefit pension plan	Amount of future benefit is **defined**, based on the plan's formula	Amount of the future obligation, based on the plan's formula, must be estimated in the **current** period.	Companies **pre-fund** by contributing funds to a pension trust.
Other post-employment benefits	Amount of future benefit depends on plan specifications and type of benefit.	Eventual benefits are specified. The amount of the future obligation must be estimated in the current	Typically **do not pre-fund**

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net liability /asset = Pension obligation (1) – Fair value of plan asset (2)

(1) Pension obligation calculation

	Pension obligation - Beginning
+	Current service cost
+	Interest cost
+	Past service cost
+	Actuarial losses
-	Actuarial gains
-	Benefits paid
	Pension obligation - Ending

- **Current service cost:** present value of benefits earned by the employees during the current period.
- **Interest cost:** the increase in the obligation due to the passage of time (time value of money). Company does not pay until employees retire => need charging time value of money
- **Past service cost:** retroactive benefits awarded to employees when a plan is initiated or amended.
- **Changes in actuarial assumptions (could be gain or loss):** the gains and losses that result from changes in variables such as mortality, employee turnover, retirement age, and the discount rate.
- **Benefits paid:** amount paid by the company to employees => decrease the pension obligation

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net liability /asset = Pension obligation (1) – Fair value of plan asset (2)

A related (but seldom used) term is the **Annual Unit Credit (benefit)**
Annual Unit Credit (benefit) = value at retirement/number of years of service.
The value at retirement is the PV of the future benefits at retirement date, while PBO under method 1 is the present value at the end of the current fiscal year.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net liability /asset = Pension obligation (1) – Fair value of plan asset (2)

(2) Fair value of net asset

Fair value at the beginning of the year
\+ Contributions
\+ Actual return
\- Benefit paid

Fair value at the end of the year

- **Contribution:** amount contributed/made by employer to the plan;
- **Actual return:** positive actual dollar return earned on plan assets;
- **Benefits paid:** amount paid by the company to employees => decrease the value of asset.

Effects on the balance sheet:

Funded status = ***Fair value of plan asset – Pension obligation***

If Obligation > Fair value assets:

Plan is underfunded → Negative funded status → Net pension liability.

If Obligation < Fair value assets:

Plan is overfunded → Positive funded status → Net pension asset.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net liability /asset = Pension obligation (1) – Fair value of plan asset (2)

Example

Question: John McElwain was hired on January 1, 2016, as the only employee of Transfer Trucking, Inc., and is eligible to participate in the company's defined-benefit pension plan. Under the plan, he is promised an annual payment of 2% of his final annual salary for each year of service. The pension benefit will be paid at the end of each year, beginning one year after retirement. McElwain's starting annual salary is $50,000.

In order to calculate the PBO at the end of the first year, we will assume the following:

- The **discount rate** is 8%.
- McElwain's salary will increase by 4% per year (this is called the **rate of compensation growth**).
- McElwain will work for 25 years.
- McElwain will live for 15 years after retirement and receive 15 annual pension benefit payments.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net liability /asset = Pension obligation (1) – Fair value of plan asset (2)

Beginning balance = 0
Current service cost

- Based on a starting salary of $50,000 in 2016 and 4% annual pay increases over 24 years, McElwain's salary at retirement will be:
- **50,000 × (1 + 0.04)^{24} = $128,165.21.** (If McElwain works for 25 years, he will receive 24 pay increases.)
- He will be entitled to an annual end-of-year pension payment equal to 2% of his final salary for each year of service. Thus, at the end of one year of service, McElwain's benefit is: **$128,165.21 × 2% × 1 year = $2,563.30 per year** from retirement until death
- Assuming he lives 15 years past retirement, the present value of the payments on the retirement date (2040) is **$21,940.55 (PV of 15 year annuity of $2,563.30 at PMT = – 2,563.30; N = 15; I/Y = 8; FV = 0; CPT PV → 21,940.55)**
- At the end of his first year of employment (2016), the present value of the annuity that begins in 24 years **is $3,460.01 ($21,940.55 discounted at 8% for 24 years)**.

No interest cost, past service cost, actuarial changes and benefit paid
=> Therefore, the PBO (FV of plan asset in the first year if employer contribute) at the end of 2016 **is $3,460.01.**

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Calculate PBO using Annual Unit Credit method

Example

Question: Continuing with John McElwain's example, calculate the PBO for the first years of his employment, and in the penultimate and fiscal year of his employment using the annual unit credit method. Ignore the possibility that McElwain may leave the company before his retirement date (in practice, this would be factored into the actuarial computations).
Given: PBO at the start of year 24 (2044) was $432,641.11.

Answer:

Salary at retirement = $128,165.21

Step 1: Calculate periodic annuity payment post-retirement (i.e., after working 25 years).
Periodic annuity payment = final salary × benefit formula × years enrolled in the plan = $128,165.21 × 0.02 × 25 = $64,082.60

Step 2: Compute the present value of the annuity at the retirement date.
N = 15 payment received post-retirement, PMT = $64,082.60, I/Y = 8%
-> CPT PV = $548,513.68

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Calculate PBO using Annual Unit Credit method

Example

Step 3: Calculate annual unit credit.

Annual unit credit = present value of annuity at the retirement date/years enrolled in the plan = \$548,513.68/25 = \$21,940.55

Step 4: Compute interest cost, current service cost, and PBO.

- Interest cost = opening PBO × discount rate

=> Interest cost year 1 = \$0 × 0.08 = \$0

- Current service cost is computed as the present value of the unit credit awarded during the year, discounted from the retirement date to the end of the current period.

=> Current service cost year 1 = \$21,940.55/ $(1.08)^{24}$ = \$3,460.01

=> PBO at end of year 1 = \$3,460.01

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net pension liability (asset)/ effects for balance sheet

Example

Question: The information below relates to the DB pension plan of two hypothetical companies as of December 31, 2010:

Company	Pension obligation	FV of plan asset
A	$8,612	$6,775
B	$3,163	$3,775

Calculate the amount that each company would report as net pension asset or liability on its 2010 balance sheet.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Net pension liability (asset)/ effects for balance sheet

Conclusion (effects on the balance sheet)

Funded status = Fair value of plan asset – Pension obligation

- If pension obligation > fair value of plan assets:

Plan is underfunded ➡ Negative funded status ➡ Net pension liability.

- If pension obligation < fair value of plan assets:

Plan is overfunded ➡ Positive funded status ➡ Net pension asset. (*)

(*) when the plan is overfunded (has a surplus), the amount of asset that can be reported is *the lower of the surplus and the **asset ceiling***.

Asset ceiling is the present value of future economic benefits, such as refunds from the plan or reductions of future contributions.

Answer:

A Company

- Step 1: Compare pension obligation and fair value of plan assets

Pension obligation = $8,613 > $6,775 = fair value of plan assets

- Step 2: Conclude accounting treatment

⇒ Plan is underfunded => net pension liability = 8,613 – 6,775 = $1,837

B Company (same as A Company)

⇒ Plan is overfunded => net pension asset = 3,775 – 3,163 = $615

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Definition

- **Periodic pension cost** is **the total cost** related to a company's DB pension plan for a given period. Various components of this total cost **can be recognized on the profit and loss (P&L)** or **in other comprehensive income (OCI)** under IFRS and U.S. GAAP.
- **Periodic pension expense** (also known as **pension cost reported in P&L**) refers to the components of periodic pension cost that **are recognized on the P&L** (not OCI).

Calculating period pension cost

Approach 1: Adjusting the change in net pension liability (Asset) for employer contributions

Net periodic pension cost = Ending net pension liability - Beginning net pension liability + Employer contributions

Approach 2: Aggregating periodic components of cost

Net periodic pension cost= Ending net pension liability - Beginning net pension liability + Employer contributions

Ending net pension liability – Beginning pension liability = changes in the pension obligation (1) and/or changes in the fair value of plan assets (2).

(1) Changes in the pension obligation:

- *Current service costs*
- *Interest costs*
- *Past service costs*
- *Actuarial losses (gains)*
- *Benefits paid to employees.*

(2) Changes in the fair value of plan assets:

- *The actual return on plan assets.*
- *Benefits paid to employees.*
- *Employer contributions.*

Periodic pension cost = Current service costs+ Interest costs+ Past service costs + Actuarial losses - Actuarial gains- Actual return on plan assets

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

APPROACH 1

Net periodic pension cost

= Employer contributions – (Ending funded status – Beginning funded status)

= Employer contributions + Ending net pension liability - Beginning net pension liability

- At inception, the net pension liability equals $0 (pension obligation= $0; fair value of plan assets = $0).
- Over Year 1 service costs equal $200 so the pension obligation rises to $200 at the end of the year (Dr Pension cost/Cr Pension obligation).
- If the employer makes no contributions to the plan, the net pension liability would increase to $200 (pension obligation = $200; fair value of plan assets= $0) → **Net periodic pension cost = Increase in net pension liability (from $0 to $200) + Employer contributions ($0) = $200.**
- If the employer contributes $200 to the plan, the net pension liability would remain unchanged at $0 (pension obligation = $200; fair value of plan assets = $200) → **Net periodic pension cost = The net pension liability has remained the same (from $0 to $0) + Employer contribution ($200).**

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

APPROACH 2

<u>Question:</u> Compute periodic pension cost given that there are no actuarial gains or losses for the period.

Reconciliation of Beginning and Ending Pension Obligation

Beginning pension obligation: $2,225

+ Current service costs: $845

+ Interest cost: $120

+ Past service costs: $635

- Benefits paid: $170

Ending pension obligation: $3,655

Reconciliation of Beginning and Ending Fair Value of Plan Assets

Beginning fair value of plan assets: $1,850

+ Actual return on plan assets: $250

+ Employer contributions: $580

- Benefits paid: $170

Ending fair value of plan assets: $2,510

Periodic pension cost = Current service costs + Interest costs + Past service costs + Actuarial losses - Actuarial gains - Actual return on plan assets

Periodic pension cost = Current service costs + Interest cost + Past service costs - Actual return on plan assets
= 845 + 120 + 635 - 250 = $1,350

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

US GAAP vs IFRS treatment of periodic pension cost

IFRS Component	IFRS recognition
Service costs	Recognized in **P&L**.
Net interest income/ Expense	Recognized in **P&L** (beginning funded status − prior service cost) × discount rate
Actuarial gains and losses	Recognized in **OCI** and not subsequently amortized to P&L. • Actuarial gains and losses = Changes in a company's pension obligation arising from changes in actuarial assumptions.

[LOS 2.d] Explain how post-employment benefits affect the financial statements

US GAAP vs IFRS treatment of periodic pension cost

US GAAP components	US GAAP recognition
Current service costs	Recognized in **P&L**.
Past service costs	Recognized in **OCI** and **subsequently amortized to P&L** over the service life of employees.
Interest cost	Recognized in **P&L**. (beginning PBO + prior service cost) × discount rate
Expected return on plan assets	Recognized in **P&L**. (Plan assets x expected return rate)
Actuarial gains and losses including differences between the actual and expected returns on plan assets	• Recognized immediately in P&L or, more commonly, recognized in OCI and **subsequently amortized to P&L using the corridor or faster recognition method** • Difference between expected and actual return on assets = Actual return - (Plan assets x Expected return). • Actuarial gains and losses =Changes in a company's pension obligation arising from changes in actuarial assumptions.

Student's notes

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

Defined benefit pension costs recognition

Question

Jacklyn King has been asked to do some accounting for Alexeeff Corp.'s pension plan. Alexeeff reports under U.S. GAAP. At the beginning of the period, the PBO was $12 million, and the fair market value of plan assets totaled $8 million. The discount rate is 9%, expected return on plan assets is $0.96 million, and the anticipated compensation growth rate is 4%.

At the end of the period, it was determined that the actual return on assets was 14%, plan assets equaled $9 million, and the service cost for the year was $0.9 million. Ignore amortization of unrecognized prior service costs and actuarial gains and losses. Periodic pension cost reported in the income statement for the year is closest to:

A. $0.72 million.

B. $0.86 million.

C. $1.02 million.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

Defined benefit pension costs recognition

Answer: C

Periodic pension cost in P&L = current service cost + interest cost – expected

Return on assets

- ***Current service cost*** = $0.90 (given)
- ***Interest cost*** = PBO at the beginning of the period × discount rate = $12 × 0.09 = $1.08
- ***Expected return on plan assets*** = $0.96 (given)

Conclusion:

Periodic pension cost in P&L = $0.90 + $1.08 – $0.96 = $1.02 million

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Corridor approach (US GAAP only)

The beginning balance of actuarial gains and losses

Max (**10%** of beginning plan assets, **10%** of beginning PBO)

The corridor approach is applied: The excess amount is amortized as a component of periodic pension cost in P&L over the remaining service life of the employees.

EXAMPLE: The following information relates to Alpha Corp., which follows U.S. GAAP:
Beginning balance of defined benefit obligation= $3,750,000
Beginning balance of fair value of plan assets= $4,200,000
Beginning balance of cumulative unrecognized actuarial losses = $450,000
Expected average remaining working lives of plan employees= 10 years
Estimate the amount of unrecognized actuarial losses that should be recognized in P&L for the current period.

ANSWER: 10% of beginning balance of defined benefit obligation= $375,000 < 10% beginning balance of fair value of plan assets = $420,000 < Beginning balance of cumulative unrecognized actuarial losses = $450,000

⇒ Corridor = $420,000 = $450,000 - $30,000

⇒ $30,000 is amortized in P&L over 10 years = $3,000 for the current period

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Assumptions in pension calculations disclosed by the firm and effects on reported result

	Increase discount rate	Decrease compensation growth	Increase expected return on plan assets
BS Liability	↓	↓	No effect
Total periodic pension cost	↓ *	↓	No effect
Periodic pension cost in P&L	↓ *	↓	↓ **

** For mature plans, a higher discount rate might increase interest costs. In rare cases, interest cost will increase by enough to offset the decrease in the current service cost, and periodic pension cost will increase.*
*** Under U.S. GAAP only. Not applicable under IFRS.*

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Assumptions in pension calculations disclosed by the firm and effects on reported result

Example

Question:
You have just been hired as the controller at Vincent, Inc. Vincent has a defined-benefit retirement plan for its employees. The firm has a relatively young workforce with a low percentage of retirees. Your first task is to analyze the effects of changing assumptions on different variables used to calculate certain pension amounts.
All else equal, a decrease in the discount rate will most likely have what impact on the funded status?

Answer:
A decrease in the discount rate will increase the PBO. A higher PBO lowers the funded status (plan assets – PBO).

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Several adjustments made for analytical purposes

Comparative financial analysis using published financial statements can be difficult when there are differences between firms' accounting treatments for pensions. Adjustments include:

- Gross vs. net pension assets/liabilities (netting pension assets versus pension obligation or not)
- Differences in assumptions used (discount rate, compensation growth, expected return on plan assets)
- Differences between IFRS and U.S. GAAP in recognizing total periodic pension cost (in income statement vs. OCI)
- Differences due to classification in the income statement (operating or non operating expenses)

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Contribution (cash) > **Obligation (noncash expense)** ⇒ **Increase CFO and decrease CFF** *(similar to an excess principal payment on a loan)*

Contribution (cash) < **Obligation (noncash expense)** ⇒ **Decrease CFO and increase CFF**

If the difference between cash flow and total periodic pension cost is material
⟹ *reclassifying the difference from operating activities to financing activities in the cash flow statement.*

Student's notes

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

Question:
Bhaskar Thakur is analyzing the financial statements of Box Car (BC) Inc., a manufacturer of equipment for the railroad industry. Thakur notices that BC has a defined benefit pension plan in place. He finds out that the company made a €340 million contribution to the plan during the year. He collects the following additional information about the company:

- Funded status (beginning of the year) €2,530
- Funded status (end of the year) €2,180

BC reported a net income during the year of €812 million. Cash flow from operating activities and financing activities were reported as €948 million and €112 million respectively. BC's tax rate was 40% during the year.
Compute:
1. The total periodic pension cost during the year.
2. Cash flow from operating and financing activities after making appropriate adjustments.

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

Net periodic pension cost
= employer contributions − (ending funded status – beginning funded status)
Effects on cash flow
- Contribution < periodic pension cost => decrease cash flow from operating activity and
- Those cash will inflow in financing activities

Answer:

1. **The total periodic pension cost** during the year:
= employer contributions – (ending funded status – beginning funded status)
= 340 – (2,180 – 2,530) = $690 million
2. Cash flow from operating and financing activities after making appropriate adjustments.
BC's contributions (in cash) < €350 million the total periodic pension cost (non cash expense).
after tax shortfall = 350(1 – 0.4) = €210 million
adjusted cash flow from operating activities = 948 – 210 = €738 million
adjusted cash flow from financing activities = 112 + 210 = €322 million

Student's notes

[LOS 2.d] Explain how post-employment benefits affect the financial statements

Example

Net periodic pension cost
= employer contributions – (ending funded status – beginning funded status)
Effects on cash flow
- Contribution < periodic pension cost => decrease cash flow from operating activity and
- Those cash will inflow in financing activities

Answer:

1. **The total periodic pension cost** during the year:
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2. Cash flow from operating and financing activities after making appropriate adjustments.

BC's contributions (in cash) < €350 million the total periodic pension cost (non cash expense).
after tax shortfall = 350(1 – 0.4) = €210 million
adjusted cash flow from operating activities = 948 – 210 = €738 million
adjusted cash flow from financing activities = 112 + 210 = €322 million

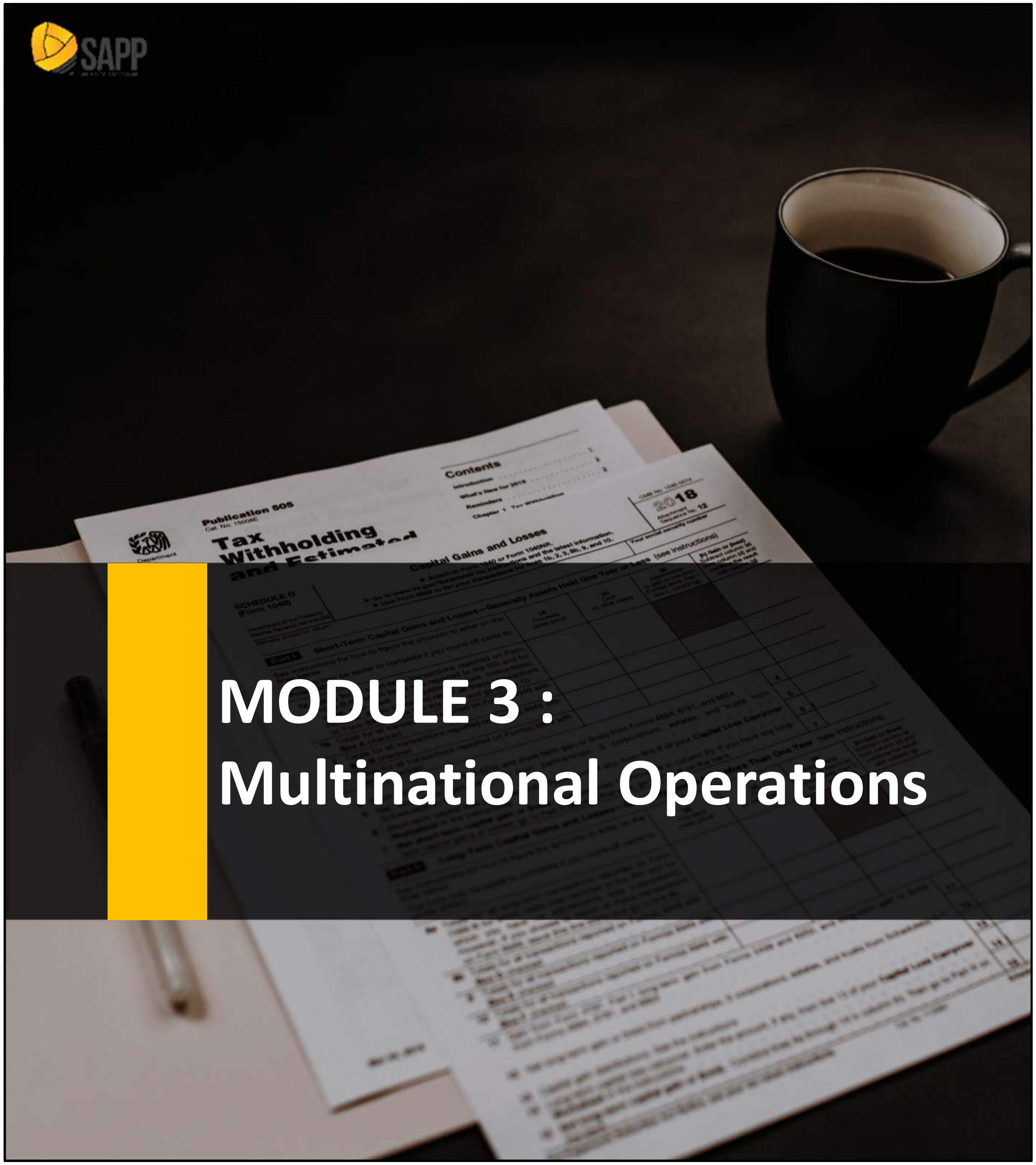
Student's notes

[LOS 2.e] Explain financial modeling and valuation considerations for post-employment benefits

Pension costs are generally embedded within a relevant expense category (e.g., SG&A, COGS), and those expenses are typically modeled as a ***percentage of revenues***.

- This is an acceptable practice because pension expenses typically vary systematically with wages/salaries.
- However, if there are changes in the plan or if the plan is closed (or frozen), separate forecasts need to be made.

- Detailed forecasts are not needed for smaller plans that are well funded or closed/frozen.
- For valuation purposes, an underfunded liability should be deducted from enterprise value.



MODULE 3 : Multinational Operations

Student's notes

LEARNING OUTCOMES

3.a.	Compare and contrast presentation in (reporting) currency, functional currency, and local currency
3.b.	Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction gains and losses
3.c.	Analyze how changes in exchange rates affect the translated sales of the subsidiary and parent company
3.d.	Compare the current rate method and the temporal method, evaluate how each affects the parent company's balance sheet and income statement, and determine which method is appropriate in various scenarios
3.e.	Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and income statement into the parent company's presentation currency
3.f.	Analyze how the current rate method and the temporal method affect financial statements and ratios

Student's notes

LEARNING OUTCOMES

3.g. Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect financial statements and ratios

3.h. Describe how multinational operations affect a company's effective tax rate

3.i. Explain how changes in the components of sales affect the sustainability of sales growth

3.j. Analyze how currency fluctuations potentially affect financial results, given a company's countries of operation

[LOS 3.a] Compare and contrast presentation in (reporting) currency, functional currency, and local currency

Currencies definition	
The local currency	The currency of the country being referred to.
The functional currency	The currency of the primary economic environment in which the entity operates, determined by management.
The presentation (reporting) currency	The currency in which the parent company prepares its financial statements.

Student's notes

Student's notes

[LOS 3.b] Describe foreign currency transaction exposure, including accounting for and disclosures about foreign currency transaction gains and losses

Accounting before Balance Sheet Date

Recognition:

- Initially recorded at the **spot exchange rate** on the transaction date.
- **Changes in the value of asset/liability** resulting from a foreign-currency transaction to be recognized as gains/losses on the income

Accounting with Intervening Balance Sheet Date

Step 1: **Unrealized** gains/losses are recognized **in the income statement**

→ **Step 2**: Once the transaction is settled. Additional gains/losses are recognized

→ **Step 3:** **Aggregating** the foreign exchange gains/losses over the two accounting periods

Analytical issues

Both IFRS and U.S. GAAP require FX G/L to be recognized on the IS, regardless of whether or not they have been realized. However, neither specifies where on the income statement.

[LOS 3.c] Analyze how changes in exchange rates affect the translated sales of the subsidiary and the parent company

Effects of foreign exposure on transactions

Transactions	Exposure	Foreign currency	
		Strengthen	Weaken
Export	**Asset** (Receivable)	Gain	Loss
Import	**Liability** (Payable)	Loss	Gain

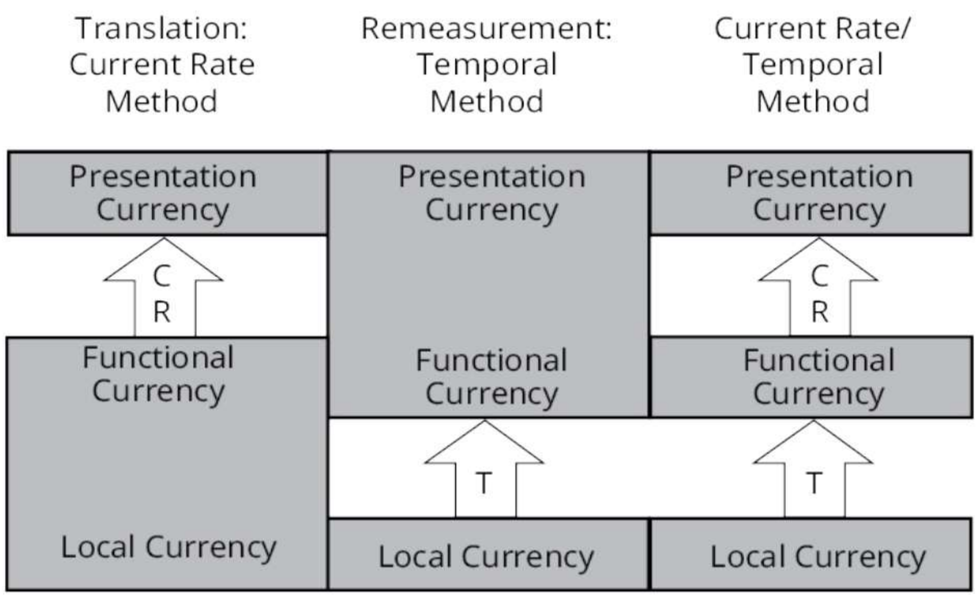
Student's notes

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Current Rate Method and Temporal Method

Remeasurement	Translation
involves converting the **local currency** into **functional currency** using the *temporal method*.	involves converting the **functional currency** into the **parent's presentation** (reporting) currency using the *current rate method*.



Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Temporal method and current rate method comparison

	Temporal Method	Current Rate Method
Monetary assets / liabilities	Current rate	Current rate
Nonmonetary assets / liabilities	Historical rates	Current rate
Common stock	Historical rates	Historical rate
Equity (taken as a whole)	Mixed	Current rate
Revenues, SG&A	Average rate	Average rate
Cost of goods sold	Historical rate	Average rate
Depreciation and amortization	Historical rate	Average rate
Net income	Mixed	Average rate
Exposure	**Net monetary assets/liabilities**	**Net assets/liabilities**
FX gain or loss	**Income statement**	**Equity**

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Balance sheet exposure under the Current method and Temporal method

		Foreign currency **Strengthen**	Foreign currency **Weaken**
Current method	**Net asset (asset > liability)**	Positive adj. (Gain)	Negative adj. (Loss)
	Net liability (asset > liability)	Negative adj. (Loss)	Positive adj. (Gain)
Temporal method	**Net monetary asset (monetary asset > monetary liability)**	Positive adj. (Gain)	Negative adj. (Loss)
	Net liability (monetary asset > monetary liability)	Negative adj. (Loss)	Positive adj. (Gain)

Current rate method:	Temporal method:
Translation start with **P/L**; final Gain/Loss is reflected on **BS**	Translation start with **B/S**; final Gain/Loss is reflected on **P/L**

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Question

FlexCo International is a U.S. company with a subsidiary, Vibrant, Inc., located in the country of Martonia. Vibrant was acquired by FlexCo on 12/31/2014. FlexCo reports its financial results in U.S. dollars. The currency of Martonia is LC.

The following exchange rates between the U.S. dollar and the LC were observed:

- December 31, 2014: $0.50 = LC1.00.
- December 31, 2015: $0.4545 = LC1.00.
- Average for 2015: $0.4762 = LC1.00.
- Historical rate for equity: $0.50 = LC1.00.
- Historical rate for PP&E and depreciation: $0.4881 = LC1.00.
- Historical rate for beginning and ending inventory $0.52/LC and $0.456/LC respectively.
- Purchases were made evenly throughout the year.

The majority of Vibrant's operational, financial, and investment decisions are made locally in Martonia, although Vibrant does rely on FlexCo for information technology expertise. Use the appropriate method to translate Vibrant's 2015 balance sheet and income statement into U.S. dollars.

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

	2014	2015		2014	2015
Cash	LC100	LC100	Account payable	400	500
A/Receivables	500	650	Current debt	100	200
Inventory	1,000	1,200	Long term debt	1,300	950
Current assets	LC1,600	LC1,950	Total liabilities	LC1,800	LC1,650
Fixed assets	800	1600	Common stock	400	400
Accumulated dep	(100)	(700)	Retained earnings	100	800 (*)
Net fixed assets	LC700	LC900	Total equity	LC500	LC1,200
Total assets	**LC2,300**	**LC2,850**	**Total liabilities and equity**	**LC2,300**	**LC2,850**

	2015		2015
Revenue	LC 5000	Other expenses	(400)
COGS	(3300)	Depreciation	(600)
Gross margin	1700	Net income	LC 700

(*) Beginning retained earning in 2015 = $50

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Current rate method

The functional currency ≠ the parent presentation currency → **current rate method**

⇒ Use **current rate** for all balance sheet accounts (except common stock)

⇒ Use **average rate** for all income statement accounts

⇒ The translation gain or loss is included in the **CTA**

2015 Translated Income Statement using the Current Rate Method

	2015 (LC)	Rate	2015 ($)
Revenue	5,000	x $0.4762	$2,381.0
COGS	(3,300)	x $0.4762	$(1,571.5)
Gross margin	1,700		$809.5
Other expenses	(400)	x $0.4762	$(190.5)
Depreciation	(600)	x $0.4762	$(285.7)
Net income	700		$333.3

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Current rate method

	2015(LC)	Rate	2015($)
Cash	100	$0.4545	$45.5
Account receivable	650	$0.4545	$295.4
Inventory	1,200	$0.4545	$545.4
Current assets	**1,950**		**$886.3**
Fixed assets	1,600	$0.4545	$727.2
Accumulated depreciation	(700)	$0.4545	$(318.2)
Net fixed assets	900		$409.0
TOTAL ASSETS	**2,850**		**$1,295.3**

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Current rate method

	2015(LC)	Rate	2015($)
Account payable	500	$0.4545	$227.2
Current debt	200	$0.4545	$90.9
Long term debt	950	$0.4545	$431.8
Total liabilities	**1,650**		**$749.9**
Common stock	400	$0.5000	$200.0
Retained earnings	800	(a) =	$383.3
CTA		(b) =	$(37.9)
Total equity	**1,200**		**$545.4**
LIABILITIES AND EQUITY	**2,850**		**$1,295.3**

(a) Ending retained earnings = $50 Beginning retained earnings + $333.3 Net income = $383.3.

(b) The CTA is a plug figure that makes the accounting equation balance: $1,295.3 assets – $749.9 liabilities – $200.0 common stock – $383.3 retained earnings = –$37.9.

Student's notes

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Temporal method

	2015(LC)	Rate	2015($)
Cash	100	$0.4545	$45.5
Account receivable	650	$0.4545	295.4
Inventory	1,200	$0.4560	545.4
Current assets	**1,950**		**888.1**
Fixed assets	1,600	$0.4881	781.0
Accumulated depreciation	(700)	$0.4881	(347.1)
Net fixed assets	LC900		$439.3
TOTAL ASSETS	**2,850**		**$1,327.4**

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Temporal method

	2015(LC)	Rate	2015($)
Account payable	500	$0.4545	227.3
Current debt	200	$0.4545	90.9
Long term debt	950	$0.4545	431.8
Total liabilities	**1,650**		**$750.0**
Common stock	400	$0.5000	200.0
Retained earnings	800	(a) =	377.4
Total equity	**1,200**		**$577.4**
LIABILITIES & EQUITY	**2,850**		**$1,327.4**

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Temporal method

	2015 (LC)	Rate	2015 ($)
Revenue	5,000	x $0.4762	2,381
COGS	(3,300)	(c) =	(1,639.5)
Gross margin	1,700		741.5
Other expenses	(400)	x $0.4762	(190.5)
Depreciation	(600)	x $0.4881	(292.9)
Income before measurement gain	700		258.1
Remeasurement gain		(d) =	69.3
Net income	700	(c) =	327.4

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

Answer: Temporal method

a) **Retained earnings** is a plug figure = $1,327.4 assets – $750.0 liabilities – $200.0 common stock = $377.4

b) **Net income** = $377.4 ending balance - $50.0 beginning balance + $0 dividends paid = $327.4

c) **COGS** = Beginning inventory + Purchases – Ending inventory

Purchases = COGS + ending Inv –beginning Inv = 3,300 + 1,200 – 1,000 = LC 3,500

→ COG in $ = (1,000 × 0.52) + (3,500 × 0.4762) – (1,200 × 0.456) = $1,639.5

d) **Remeasurement gain** is a plug figure = $327.4 net income – $258.1 income before remeasurement gain = $69.3

Student's notes

[LOS 3.d] Compare the current rate method and the temporal method

	2015($) Current rate	2015 ($) Temporal
Cash	$45.5	$45.5
Account receivable	295.4	295.4
Inventory	545.4	547.2
Current assets	**$886.3**	**$881.1**
Fixed assets	727.2	781.0
Accumulated depreciation	(318.2)	(341.7)
Net fixed assets	$409.0	439.3
TOTAL ASSETS	**$1,295.3**	**$1,327.4**
Account payable	227.2	227.3
Current debt	90.9	90.9
Long term debt	431.8	431.8
Total liabilities	**$749.9**	**750.0**
Common stock	200.0	200.0
Retained earnings	383.3	377.4
CTA	(37.9)	
Total equity	**$545.4**	**577.4**
LIABILITIES AND EQUITY	**$1,295.3**	**$1,327.4**

Student's notes

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and income statement into the parent's presentation currency

	2015($) Current rate method	2015($) Temporal method
Revenue	$2,381.0	$2,381.0
COGS	(1571.5)	(1695.5)
Gross margin	809.5	741.5
Other expenses	(190.5)	(190.5)
Depreciation	(285.7)	(292.9)
Income before measurement gain	333.3	258.1
Remeasurement gain		69.3
Net income	$333.3	$327.4

Student's notes

[LOS 3.e] Calculate the translation effects and evaluate the translation of a subsidiary's balance sheet and…

Effects of Exchange Rate Movements on Financial Statements (Summary)

	Temporal method, net liability exposure	Temporal method, net asset exposure	Current rate method
Foreign currency strengthens	Revenues: (+) Assets: (+) Liabilities: (+) Net income: (-) Equity: (-) **Translation loss**	Revenues: (+) Assets: (+) Liabilities: (+) Net income: (+) Equity : (+) **Translation gain**	Revenues: (+) Assets: (+) Liabilities: (+) Net income: (+) Equity : (+) **Positive adjustment**
Foreign currency weakens	Revenues: (-) Assets: (-) Liabilities: (-) Net income: (+) Equity: (+) **Translation gain**	Revenues: (-) Assets: (-) Liabilities: (-) Net income: (-) Equity : (-) **Translation loss**	Revenues: (-) Assets: (-) Liabilities: (-) Net income: (-) Equity: (-) **Negative adjustment**

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

Pure balance sheet ratios, pure income statement ratios, and mixed ratios

Definition

- **Pure balance sheet ratio:** the numerator and denominator are both balance sheet items
- **Pure income statement ratio:** the numerator and denominator are both balance sheet items
- **Mixed ratio**: one of the inputs comes from the balance sheet and the other from the income statement

The effects of translation methods on financial ratios

Under the current rate method

- Pure balance sheet and pure income statement ratios will be the same (all income statement items are translated at the average rate, all balance items are translated at current rate)
- If the foreign currency is depreciating, translated mixed ratios (with an income statement item as numerator) will be larger than the original ratio.
- If the foreign currency is appreciating, translated mixed ratios (with an income statement as numerator) will be smaller than the original ratio

Under the temporal method

Income statement items, balance sheet items translated at different rate => more considerations need to be taken when analyzing ratios

Student's notes

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

Determine whether the foreign currency is **appreciating or depreciating**

↓

Determine which rate is used to convert the numerator under both methods & whether the **numerator of the ratio** will be the same, larger, or smaller under the temporal method versus the current rate method.

↓

Determine which rate is used to convert the denominator under both methods & whether **the denominator of the ratio** will be the same, larger, or smaller under the temporal method versus the current rate method.

↓

Determine whether the ratio **will increase, decrease, or stay the same** based on the direction of change in the numerator and the denominator

Student's notes

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

Question

Which of the following ratios is unaffected by the choice between the current rate method and the temporal method ?

A. Quick ratio

B. Accounts payable turnover

C. Current ratio

Answer

A is correct

$$\text{Quick ratio} = \frac{\text{Cash + accounts receivables}}{\text{Current liabilities}}$$

$$= \frac{\text{Cash + cash equivalents + accounts receivable}}{\text{Accruals + accounts payable + notes payable}}$$

Numerator: translated at current rate under both methods

Denominator: translated at current rate under both methods

B is incorrect because Accounts payable turnover includes purchases (the same as inventory)

C is incorrect because Current ratio includes inventory (translated at current rate under current rate method, translated at historical rate under the temporal method)

Student's notes

[LOS 3.f] Analyze how the current rate method and the temporal method affect financial statements and ratios

Hyperinflation overview

Hyperinflation
(FASB: 100% inflation rate over 3 years – 26% compound rate each year)

Much lower assets and liabilities after translation **using the current rate** method

The **real value** of the nonmonetary assets and liabilities is typically **not affected** by hyperinflation

Adjusting the **nonmonetary asset and liabilities** for inflation

US GAAP (not allowed)	**IFRS (**allowed)

Accounting for subsidiaries operating under hyperinflation:

IFRS: the foreign currency financial statements are restated for inflation then translated using the current exchange rate

US GAAP: the functional currency is considered to be the parent's presentation currency; thus, the temporal method is used to remeasure the financial statements.

Student's notes

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary economies affect financial statements and ratios

Example

In a hyperinflationary economy, translation under the current rate method will most likely result in relatively

A) high balance sheet values for long term assets

B) high translation gains.

C) low balance sheet values for long term liabilities.

Answer: In a hyperinflationary economy, translation under the current rate method will most likely result in relatively low values for assets and liabilities. Translation losses will occur.
C is correct

Which of the following asset or liability values is likely to be the most understated in a hyperinflationary economy if translation occurs under the current rate method?

A) Dividends payable.

B) A plant purchased several years ago.

C) Accounts receivable.

Answer: B The accounts receivable and dividends payable will each have book values that are closer to their market values than a plant purchased many years ago.
B is correct

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary…

Restating for inflation process (IFRS)

Nonmonetary assets/liabilities are restated for inflation **using a price index → not necessary** to restate monetary assets/liabilities)

↓

The components of **shareholders' equity** are restated by applying the **change in the price index** from the beginning of the period or the date of contribution if later.

↓

The **income statement** items are restated **by multiplying by the change in the price index** from the date the transactions occur.

↓

All items are then translated into the parent's presentation currency using the **current exchange rate**.

↓

The net purchasing power gain or loss is recognized in the income statement based on the net monetary asset or liability exposure.

Student's notes

Student's notes

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary…

Restating for inflation process (IFRS) - Example

	2014	2015
Cash	5,000	8,000
Supplies	25,000	25,000
Total assets	30,000	33,000
Accounts payable	20,000	20,000
Common stock	10,000	10,000
Retained earnings	0	3,000
Liabilities and equity	30,000	33,000
Revenue		15,000
Expenses		(12,000)
Net income		3,000

Question

Imagine that a foreign subsidiary was created on December 31, 2014. The LC is the currency of the country where the foreign subsidiary is located. The subsidiary's balance sheets for 2014 and 2015, and income statement for the year-ended 2015, are shown in the following

Also, use the following price indices:

- December 31, 2014 100
- December 31, 2015 150
- Average for 2015 125

Prepare an inflation adjusted balance sheet and income statement for 2015

Student's notes

[LOS 3.g] Analyze how alternative translation methods for subsidiaries operating in hyperinflationary…

Restating for inflation process (IFRS) - Example

In LCs	2015	Adjustment factor	Inflation adjusted
Cash	8,000		8,000
Supplies	25,000	150/100	37,500
Total assets	33,000		45,000
Accounts payable	20,000		20,000
Common stock	10,000	150/100	15,000
Retained earnings	3,000		10,500
Liabilities and equity	33,000		45,500
Revenue	15,000	150/125	18,000
Expenses	(12,000)	150/125	(14,400)
Net purchasing power gaine			6,900
			10,500

[LOS 3.h] Describe how multinational operations affect a company's effective tax rate

Tax rate for multinational companies

Statutory tax rate		Effective tax rate
Provided by the tax code of the home country.	≠	The tax expense in the income statement divided by pretax profit.

Reconcilation required

- Changes in the mix of profits from different countries (with varying tax rates)
- Changes in the tax rates

→ Changes in effective tax rate on account of foreign operations

Student's notes

Student's notes

[LOS 3.h] Describe how multinational operations affect a company's effective tax rate

Example

Question

The reconciliation between the statutory tax rate and effective tax rate for two companies (Amco and Bianco) for the year 2017 is provided.

Item	Amco	Bianco
Statutory tax rate	25.0%	30.0%
Effect of disallowed expenses	3.0%	1.0%
Effect of exempt income	(2.0%)	(0.5%)
Effect of taxes in foreign jurisdictions	3.4%	(1.2%)
Effect of recognition of prior losses	(0.8%)	(3.0%)
Effective tax rate	28.6%	26.3%

1. Which company benefited from the lowering of tax expense on account of its foreign operations?
2. If the mix of foreign operations for both companies is expected to increase over time, which company is most likely to report lower effective tax rate in the future? Assume that the statutory tax rates do not change.

Answer

1. The effect of foreign operations resulted in an increase in effective tax rate for Amco by 3.4% and a decrease for Bianco by 1.2%. Hence, Bianco benefited from its foreign operations in reducing its effective tax rate and tax expense.
2. If the mix of foreign operations increases, assuming no change to the tax rate, Bianco would see its effective tax rate decrease further and Amco would see its effective tax rate increase.

Student's notes

[LOS 3.i] Explain how changes in the components of sales affect the sustainability of sales growth

Financial statements for multinational companies report sales **denominated in the reporting currency**

Changes in currency values affect the value of translated sales

Sales growth owing to an increase in volumes or prices is considered more sustainable than currency appreciation

% Change in Sales Excluding Currency Effect
= % Change in Sales – Impact of currency

Student's notes

[LOS 3.i] Explain how changes in the components of sales affect the sustainability of sales growth

Question

BCN, Inc., an American multinational, reported the following in its MD&A for the year 2017:

Region	% Change in Sales	Impact of Currency	% Change in Sales Excluding Currency Effect
North America	3%	-2%	5%
Asia	2%	0%	2%
Europe	2%	3%	-1%

Comment on the growth rates observed in different markets for BCN ?

Answer

- **In North America**, BCN experienced highest growth in North America. Excluding the loss in translation (due to depreciation of the foreign currency in which sales are denominated) for North America ex-U.S., the revenue growth would have been 5% due to an increase in volumes and/or price.
- **In Europe**, the revenue growth is 2%, but that was due to appreciation of the currencies in which sales occurred. Excluding the currency effect, sales were down in Europe by 1%.
- **In Asia**, the revenue growth of 2% is solely due to price/volume changes.

Student's notes

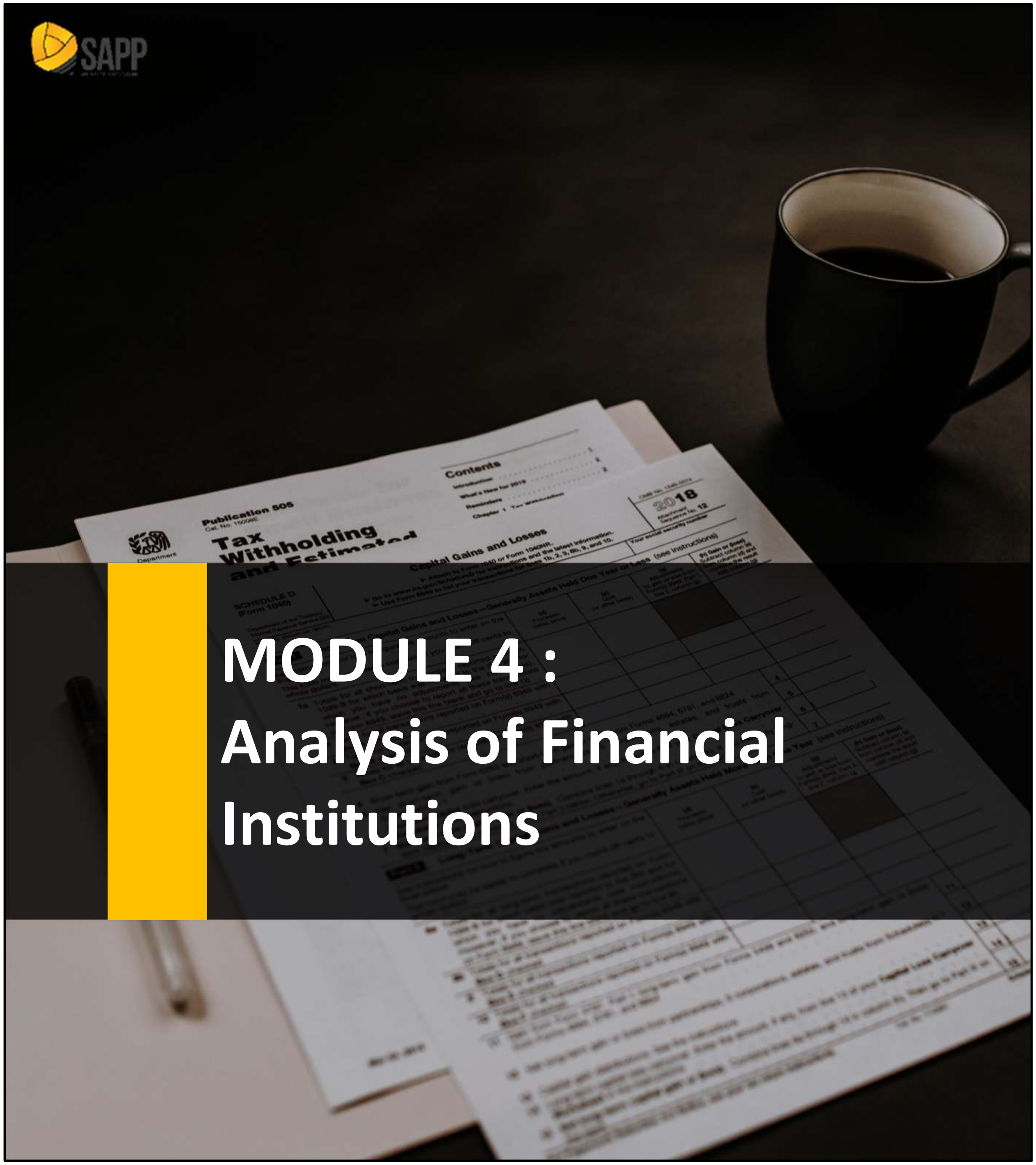
[LOS 3.j] Analyze how currency fluctuations potentially affect financial results, given a company's countries of operation

Example: Walmart's foreign exchange risk management practice

An excerpt from Walmart's 2015 Annual Report:

We are exposed to fluctuations in foreign currency exchange rates as a result of our net investments and operations in countries other than the U.S. For fiscal 2015, movements in currency exchange rates and the related impact on the translation of the balance sheets of the Company's subsidiaries in Canada, the United Kingdom, Japan, Mexico, and Chile were the primary cause of the $3.6 billion net loss in the currency translation and other category of accumulated other comprehensive income (loss). We hedge a portion of our foreign currency risk by entering into currency swaps and designating certain foreign currency– denominated long-term debt as net investment hedges.

We hold currency swaps to hedge the currency exchange component of our net investments and also to hedge the currency exchange rate fluctuation exposure associated with the forecasted payments of principal and interest of non-U.S.-denominated debt. The aggregate fair value of these swaps was in a liability position of $110 million at January 31, 2015, and in an asset position of $550 million at January 31, 2014. The change in the fair value of these swaps was due to fluctuations in currency exchange rates, primarily the strengthening of the U.S. dollar relative to other currencies in the latter half of fiscal 2015. A hypothetical 10% increase or decrease in the currency exchange rates underlying these swaps from the market rate at January 31, 2015, would have resulted in a loss or gain in the value of the swaps of $435 million. A hypothetical 10% change in interest rates underlying these swaps from the market rates in effect at January 31, 2015, would have resulted in a loss or gain in value of the swaps of $20 million.



MODULE 4 : Analysis of Financial Institutions

Student's notes

LEARNING OUTCOMES

4.a. Describe how financial institutions differ from other companies

4.b. Describe key aspects of financial regulations of financial institutions

4.c. Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations

4.d. Analyze a bank based on financial statements and other factors

4.e. Describe other factors to consider in analyzing a bank

4.f. Describe key ratios and other factors to consider in analyzing an insurance company

Student's notes

Student's notes

[LOS 4.a] Describe how financial institutions differ from other companies

FINANCIAL INSTITUTION DIFFERENCES	**Systematic importance**	• Financial institutions are necessary for the smooth functioning and overall health of the economy • Potential contagion effect can happen due to **interlinkage** between financial institutions
	Level of regulation	Financial institutions are **highly regulated** **Example:** minimum capital requirements, minimum liquidity requirements, and limits on risk taking,...
	Assets	• **Financial institutions**: mostly financial assets **reported at fair value** • **Other companies**: primarily own tangible assets reported at depreciated historic cost.

Student's notes

[LOS 4.b] Describe key aspects of financial regulations of financial institutions

Basel III Framework

Stable funding relative to a bank's liquidity needs over a one-year time horizon required	**Minimum required capital** for a bank is based on the risk of the bank's assets.	A bank should hold enough **liquid assets** to meet demands under a 30-day liquidity stress scenario.

Other international organizations to coordinate regulations	Functions
Financial Stability Board"	Coordinate actions of participating jurisdictions in identifying and managing systemic risks.
International Association of Deposit Insurers	Improve the effectiveness of deposit insurance systems.
International Organization of Securities Commissions (IOSCO)	Promote fair and efficient security markets.
International Association of Insurance Supervisors (IAIS)	Improve supervision of the insurance industry.

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing a bank, including key ratios and its limitations; Analyze a bank based on financial statements and other factors

CAMELS overview

C	=	Capital adequacy (1)
A	=	Asset quality (2)
M	=	Management (3)
E	=	Earnings (4)
L	=	Liquidity (5)
S	=	Sensitivity (6)

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

(1) C – capital adequacy (calculated based on risk weighted asset)

Total capital of a bank		
Common Equity Tier 1 capital	Other Tier 1 capital	Tier 2 capital
Common stock, +Additional paid-in capital +Retained earnings +OCI - Intangibles & deferred tax assets.	Subordinated instruments with <u>no specified maturity</u> and <u>no contractual dividends</u>	Subordinated instruments with original maturity of more than five years.

Total assets value	
Loans	**Investment in Securities**

BASEL III requirements

Common Tier 1 capital	≥ 4.5%	of risk weighted assets
Total Tier 1 capital	≥ 6%	of risk weighted assets
Total capital	≥ 8%	of risk weighted assets

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

C – capital adequacy (continued)

Bank Asset Classification System (According to Basel I)	
0 % Risk	Cash, government debt, central bank debt, and the debt of governmental departments and organizations
10 % Risk	Central bank debt of countries with high inflation in the recent past
20 % Risk	Development debts, OECD bank debt, non OECD bank debt under one year of maturity, and non OECD public debt sector
50 % Risk	Residential mortgage
100 % Risk	Private sector debt, non OECD bank debt with maturity over a year, real estate, plant and equipment, and capital instruments issued at other banks

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

C – Capital Adequacy (example)

Question

Mega Bank's RWA for 20X7 and 20X8 were $4,700,000 and $5,175,000, respectively. An analysis of bank's capital for the two years reveals the following information.

1. Determine the Common Equity Tier 1 capital ratio, Tier 1 capital ratio and total capital ratio for both years.
2. 2. Comment on capital adequacy for Mega Bank.

	20X7 (000)	20X8 (000)
Common Equity Tier 1	$2000	$190
Subordinated debt, no maturity, no dividends/interests	$85	$95
Subordinated debt, maturity 5+ year	$110	$120

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

C – Capital Adequacy (example)

Answer

Mega Bank's capital ratios as a percentage of its RWA for the two years are shown in the following table.

	20X7 (% of RWA)	20X8 (% of RWA)
Common Equity Tier 1	4.26%	3.67%
Subordinated debt, no maturity, no dividends/interest	1.81%	1.84%
Subordinated debt, maturity 5+ year	2.34%	2.32%
Total capital	8.40%	7.83%

Total Tier 1 ratio for 20X7 = 4.26% + 1.81% = 6.07%

Total Tier 1 ratio for 20X8 = 3.67% + 1.84% = 5.51%

- *Common Equity Tier 1 ratio* < Basel III guideline of 4.5%. Furthermore, it has worsened from 20X7 to 20X8 → most important component of capital → CONCERN
- *Total Tier 1 ratio* was above the Basel III guideline of 6% in 20X7, but has fallen below the guideline for 20X8 → CONCERN
- *Total capital ratio* is above the Basel III guideline of 8% for 20X7 but again has fallen short in 20X8 → CONCERN

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

A – Asset Quality

Loans	Investment in securities

Bank asset

Generating assets	Managing assets	Controlling overall risk

↓

Asset quality

↓

Key consideration for analyzing asset quality:

- Credit risk associated with a bank's loan portfolio
- Credit risk associated with a bank's investments.
- Counterparty credit risk associated with trading activities.
- Potential credit risk linked to off-balance-sheet obligations such as guarantees and letters of credit.
- Diversification of credit risk exposure across a bank's financial assets.
- Liquidity and other factors that affect asset value and marketability.

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

A – Asset Quality (Example)

Assets	2017	2016
Cash and balances at central banks	171,082	102,353
Items in the course of collection from other banks	2,153	1,467
Trading portfolio assets	113,760	80,240
Financial assets designated at fair value	116,281	78,608
Derivative financial instruments	137,669	246,626
Financial investments	58,916	63,317
Loans and advances to banks	35,663	43,251
Loans and advances to customers	465,552	492,784
Reverse repurchase agreements	12,546	13,454
Assets included in disposal groups classified as held for sale	1,193	71,454
Other assets	18,433	19,572
Total Assets	1,133,248	1,213,126

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and...

A – Asset Quality (Example - continued)

Question

- 1. Considering the most liquid assets (cash and balances at central banks and items in the course of collection from other banks), how did XYZ's balance sheet liquidity change from 2016 to 2017?
- 2. What is the proportion of investments to total assets in 2016 and 2017?
- 3. What is the proportion of loans to total assets in 2016 and 2017?

Answer

- Balance sheet liquidity in 2016=(102,353+ 1,467)/11,213,126 =8.6%
- Balance sheet liquidity in 2017=(171,082+2,153)/11,133,248 = 15.3%

→ XYZ's balance sheet liquidity increased significantly in 2017.

XYZ's investments comprised trading portfolio assets, financial assets designated at fair value, derivative financial instruments, and financial investments.

- Proportion of investments/Total assets in 2016
  =(80,240+ 78,608+ 246,626+63,317)/1,213,126 =38.6%
- Proportion of investments/Total assets in 2017
  =(113,760+ 116,281 + 137,669+58,916)/1,133,248 =37.6%

XYZ's loans comprised loans and advances to banks, loans and advances to customers, and reverse repurchase agreements.

- Loans/Total assets in 2016 =(43,251 +492,784+ 13,454)/1,213, 126 =45.3%
- Loans/Total assets in 2017 =(35,663 +465,552+ 12.546)/l, 133,248 =45.3%

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

M – Management & E - Earnings

Management

Measures the ability of the management team to identify and then react to financial stress, exploit profitable opportunities while also controlling the level of risks taken

Management quality ← Business strategy; Governance; Internal control

Banks earning

Net interest income	Net fee and commission income	Net trading income

- Net interest income and net fee and commission income are more sustainable
- A bank with less volatility in net interest income is not excessively exposed to interest rate risk
- Earnings are considered high quality if they are adequate as well as sustainable (recurring); estimates must be unbiased

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and...

Earning analysis example

Question

An analyst has extracted the following information on XYZ Bank's operating income.

1. What percentage of XYZ's total operating income was earned from (a) net interest income, (b) net fee and commission income, and (c) non trading income in 2017?

2. Comment on the trend in the bank's net interest income and total operating income.

	2017	2016	2015
Net interest income	9,845	10,537	10,608
Net fee and commission income	6,814	6,768	6,859
Net trading income	3,500	2,768	3,426
Net investment income	861	1,324	1,097
Other operating income	56	54	50
Total operating income	21,076	21,451	22,040

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

Earning analysis example - continued

Answer

- Percentage of XYZ's total operating income from net interest income

  = 9,845/21,076 =46.7%

- Percentage of XYZ's total operating income from net fee and commission income = 6,814/21,076 = 32.3%

- Percentage of XYZ's total operating income from net trading income

  = 3,500/21,076 = 16.6%

- XYZ's net interest income and total operating income have declined year-on-year from 2015 to 2017. The percentage Net interest income/Total operating income has remained relatively stable around 48% over the period.

  2015: 10,608/22,040 =48.1%

  2016: 10,537/21,451 =49.1%

  2017: 9,845/21,076 =46.7%

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

L - Liquidity

Liquidity Analysis (Basel III)

Liquidity coverage ratio (LCR)

$$LCR = \frac{\text{highly liquid assets}}{\text{expected cash outflow*}}$$

**est. 1-month liquidity in a stress scenario*

Concentration of funding

An excessively high concentration of funding indicates vulnerability to a single source of funding being withdrawn

Contractual maturity mismatch

- This compares the maturities of a bank's assets and funding sources.
- Maturity mismatch exposes a bank to liquidity risk if cash repayments from borrowers are insufficient to meet the amount required for maturing deposits.

Net stable funding ratio (NSFR)

$$NSFR = \frac{\text{Available stable funding}}{\text{Required stable funding}}$$

- **Available Stable Funding** depends on the composition and maturity of **funding sources** (<u>Example</u>: long-dated deposits are more stable than short-dated deposits)
- **Required Stable Funding** depends on the composition and maturity of **asset base**.
- Basel III sets a minimum NSFR of 100%.

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

L – Liquidity (Example)

Figure 16.1: Available Stable Funding Components

Funding Component	ASF Factor
Regulatory capital minus Tier 2 instruments maturing in a year Other capital instruments and liabilities with maturity > 1 year	100%
Stable demand deposits and term deposits (maturity < 1 year) from retail and small business customers	95%
Less-stable demand deposits and term deposits (maturity < 1 year) from retail and small business customers	90%
Funding from nonfinancial corporates (maturity < 1 year), operational deposits, funding from sovereigns, public sector (maturity < 1 year), multilateral and national development banks	50%
All other liabilities	0%

Question

Describe the liquidity risk faced by Mega Bank

	20X9	20X8
Highly liquid assets	1,250	1,100
Average monthly withdrawals	900	750
Expected monthly outflows in a stress scenario	1,300	1,050

Answer

LCR for 20X8 = (1,100/1,050) = 104.76%

LCR for 20X9 = (1,250/1,300) = 96.15%

=> For 20X9, the bank's LCR has dropped from prior year, dropping even below the minimum requirement of 100%, indicating higher liquidity risk.

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

S - Sensitivity

Market risk

Currency values	Interest rates (The most important factor)	Volatility of security prices

↓

Banks' composition of assets and liabilities

↓

Banks' earning

Interest rate risk: the result of differences in maturity, rates, and repricing frequency between the bank's assets and its liabilities.

In rising interest rate scenario → if the assets are repriced more frequently than liabilities → net interest income would increase.

For example: following the financial crisis of 2008, central banks around the world reduced short-term interest rates, allowing banks to borrow at lower rates → banks increased their duration risk (i.e., borrowed more short-term funds while lending long term).

Student's notes

[LOS 4.c,d] Explain the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and…

S – Sensitivity (Example)

Question: The following table shows XYZ Bank's sensitivity analysis on pretax net interest income for nontrading financial assets and liabilities. This metric assumes an instantaneous parallel change to interest rate forward curves. The main model assumptions are: (1) a one-year time horizon; and (2) balance sheet is held constant. How would pretax net interest income change if the interest rate forward curves shifted upward by 25 basis points?

	TOTAL
As at 31 December, 2017	
+100 basis point	76
+25 basis point	23
-25 basis point	(83)

Answer: Pretax net interest income would increase by 23 if the forward curves shifted upward by 25bps.

Student's notes

[LOS 4.e] Describe other factors to consider in analyzing a bank

Other factors to consider in analyzing a bank

Government Support: the larger the bank and more inter-linked it is → the more likely contagion effect → higher probability of implicit government support → support level is inversely related to the overall health of the bank

Bank Mission: banks may be guided by development mission in their lending → concentration of risk in asset portfolio

Government Ownership → Public ownership increases faith (implicit gov't backing) in a bank → crisis → governments became reluctant owners of struggling banks

Culture:

- Diversity of a bank's assets → narrow focus = aggressive
- Accounting restatements → unethical culture
- Management compensation → tie to banks stock → excessive risk-taking
- Speed at which a bank adjusts its loan loss provisions relative to actual loss behavior.

General factors:

- Competitive environment
- Off-balance-sheet assets and/or liabilities
- Segment & currency exposure

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Property & casualty (P&C) insurance companies

Definition

Property and casualty (P&C) insurers are companies that provide coverage on assets (e.g., house, car, etc.) and also liability insurance for accidents, injuries, and damage to other people or their belongings.

Customers
Outline desired coverage

→

P&C insurance companies
- Determine riskiness of customers
- Determine insurance premium

→

Customers
Decide whether to accept coverage and pay insurance premium

Characteristics of P&C insurance companies:
- P&C policies are generally short-term in nature
- Claims arise from unpredictable accidents
- Claims may even occur after the policy has expired

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Property & casualty insurance companies (continued)

P&C insurance companies' profitability

Prudence in underwriting

Prudence in pricing of adequate premiums for bearing risk

Diversification of risk

Insurers often reinsure some risks.
The policy period is often very short, with premiums received at the beginning of the period and invested during the float period.
Claim events are clearly defined but may take a long time to emerge.

Earnings characteristics

The property and casualty insurance business is cyclical.

"soft" pricing market	"hard" pricing market

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Ratios to analyze the profitability of P&C insurers profitability

Combined ratio = Underwriting loss ratio + Expense ratio

Underwriting loss ratio

$$\frac{\text{claim paid} + \Delta\text{loss reverse}}{\text{net premium earned}}$$

Expense ratio

$$\frac{\text{underwriting expenses including commissions}}{\text{net premium written}}$$

Other Profitability Ratios

Loss and loss adjustment expense ratio

$$\frac{\text{loss expense} + \text{loss adjustment expense}}{\text{net premiums earned}}$$

Dividends to policyholders ratio (2) =

$$\frac{\text{dividends to policy holders (shareholders)}}{\text{net premiums earned}}$$

Combined ratio after dividends (CRAD) = combined ratio + dividends to policyholders ratio

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Ratios to analyze the profitability of P&C insurers example

Question

Jia Li, CFA, is reviewing the relative performance of three P&C insurers. Li has collected the following information for the year 20X9.

Description ($ millions)	**IPCO**	**SYMCO**	**DELPHI**
Loss and loss adjustment expense	4,512	2,278	3,265
Underwriting expense	1,322	1,799	1,867
Dividends to policyholders	122	198	148
Net premiums earned	7,598	4,978	5,994
Net premiums written	7,682	5,222	6,322

Which insurer with the highest combined ratio after dividends ?

Formula: combined ratio = the loss and loss adjustment expense ratio + underwriting expense ratio = (loss expense + loss adjustment expense/ net premium earned) + (underwriting expenses/net premium earned)

Answer

Ratio	**IPCO**	**SYMCO**	**DELPHI**
Loss and loss adjustment expense ratio	59.38%	45.76%	54.47%
Underwriting expense ratio	17.21%	34.45%	29.53%
Combined ratio	76.59%	80.21%	84.00%
Dividends to policyholders ratio	1.61%	3.98%	2.47%
Combined ratio after dividends	78.2%	84.19%	86.47%

DELPHI has the highest combined ratio after dividends of 86.47%

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

P&C insurance companies' investment return, liquidity and capitalization

Investment return

- After premium income, investment return is an important source of P&C insurers 'profitability.
- Total investment return ratio = total investment income ÷ invested assets.

Liquidity

Liquidity is an important consideration for P&C insurers as they stand ready to meet their claim obligations and can be evaluated by looking at their fair value hierarchy reporting.

Capitalization

There are no global risk-based capital requirement standards for insurers.

- The EU: Solvency II standards.
- The US: NAIC

Student's notes

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Life and Health (L&H) Insurance Companies

Definition: Life and health (L&H) insurers are companies that provide coverage on the risk of life and medical expenses incurred from illness or injuries.

Customers
Outline desired coverage

→

L&H insurance companies
- Determine riskiness of customers, insurance premium
- Determine insurance premium

→

Customers
Decide whether to accept coverage and pay insurance premium

Characteristics of L&H insurance companies:
- Policies can be basic term-life
- Claims are more predictable compared to P&C insurers

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Life and Health (L&H) insurance companies analysis

Considerations in analysis of L&H insurers

- Revenue diversification
- Liquidity
- Investment returns
- Capitalization
- Earnings characteristics
  - Standard ratios
  - Industry-specific cost ratios

Industry-specific cost ratios:

1) Total benefits paid ÷ net premiums written and deposits.
2) Commissions and expenses ÷ net premiums written and deposits.

Student's notes

[LOS 4.f] Describe key ratios and other factors to consider in analyzing an insurance company

Life and Health (L&H) insurance companies analysis (continued)

Investment return

Investment returns are a key component of the insurer's profitability

Total investment return ratio = total investment income ÷ invested assets

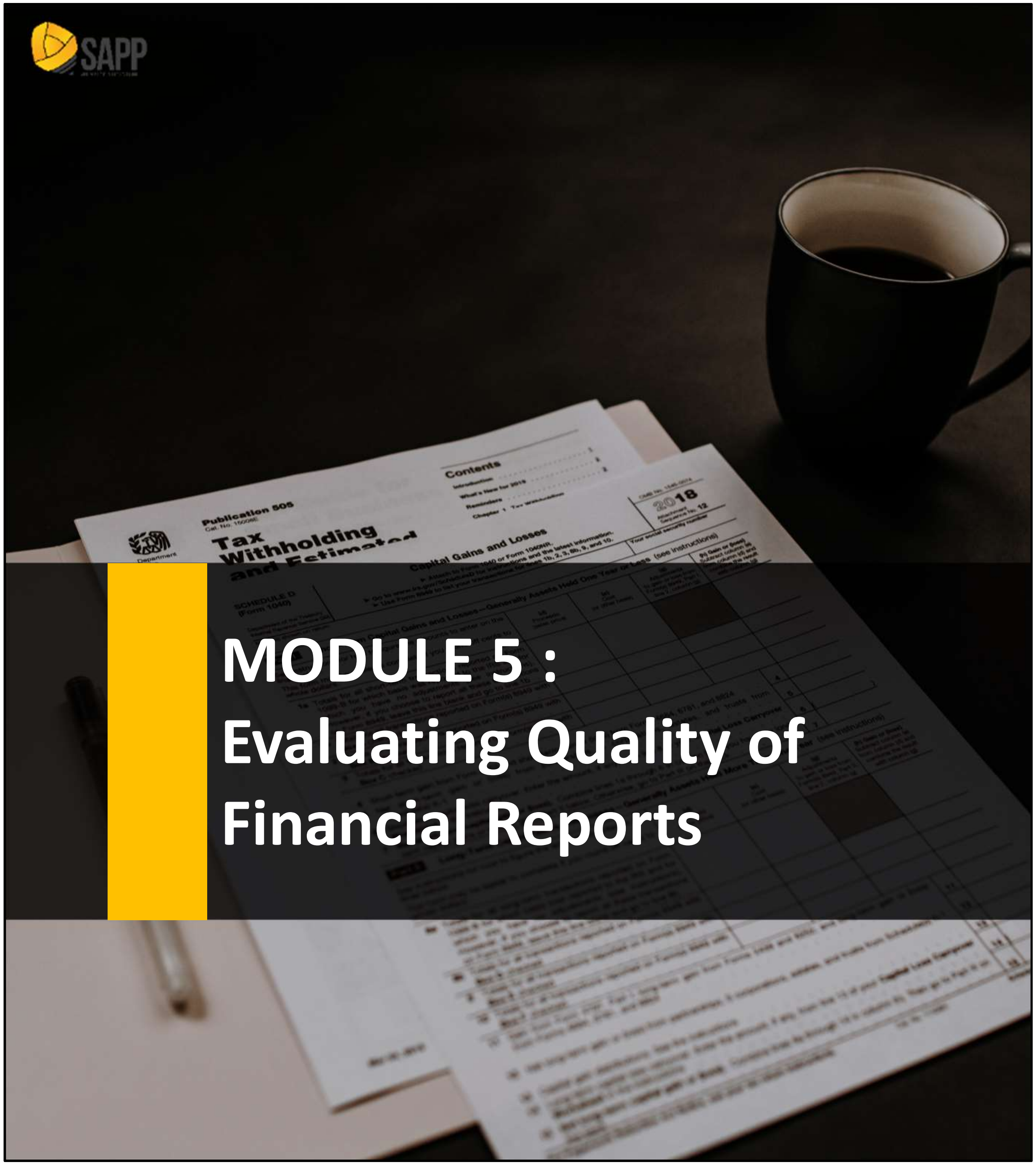
Liquidity

The liquidity needs of L&H insurers are generally fairly predictable → keeping excess liquidity is not as much of a concern for L&H insurers compared to P&C insurers → Liquidity analysis of L&H companies focus on investment portfolio

Capitalization

Similar to P&C insurers, there are no global minimum capitalization standards

Domestic regulators often do specify risk-adjusted minimum capital requirements



MODULE 5 : Evaluating Quality of Financial Reports

Student’s notes

Student's notes

LEARNING OUTCOMES

5.a. Demonstrate the use of a conceptual framework for assessing the quality of a company's financial reports

5.b. Explain potential problems that affect the quality of financial reports

5.c. Describe how to evaluate the quality of a company's financial reports

5.d. Evaluate the quality of a company's financial reports

5.e. Describe indicators of earnings quality

5.f. Describe the concept of sustainable (persistent) earnings

5.g. Explain mean reversion in earnings and how the accruals component of earnings affects the speed of mean reversion

5.h. Evaluate the earnings quality of a company

5.i. Evaluate the cash flow quality of a company

5.j. Describe indicators of balance sheet quality

LEARNING OUTCOMES

5.k. Evaluate the balance sheet quality of a company

5.l. Describe indicators of cash flow quality

5.m. Describe sources of information about risk

Student's notes

Student's notes

[LOS 5.a] Demonstrate the use of a conceptual framework for assessing the quality of a company's financial reports

The quality of financial reports		
Earnings quality		**Reporting quality**
Level of earnings	Sustainability of earning	An assessment of the information disclosed in the financial reports.

Financial reporting quality (High to low)

1. GAAP compliant and decision useful, high quality earning
2. GAAP compliant and decision useful, low quality earning
3. GAAP compliant but not decision useful (biased choices)
4. Non compliant accounting
5. Fraudulent accounting

Student's notes

[LOS 5.b] Explain potential problems that affect the quality of financial reports

Measurement and timing issues

Financial statements are **integrated** → Errors in measurement and/or timing can affect **many elements**

- **Aggressive, premature, and fictitious revenue recognition** results in overstated income and thus overstated equity. Assets, usually accounts receivable, are also overstated
- **Conservative revenue recognition**, such as deferred recognition of revenue, results in understated net income, understated equity, and understated assets.
- **Cash flow from operations** may be increased by deferring payments on payables, accelerating payments from customers, deferring purchases of inventory, and deferring other expenditures related to operations, such as maintenance and research.

Student's notes

[LOS 5.b] Explain potential problems that affect the quality of financial reports

Classification issues

Misclassification	Effect
Removing accounts receivable by selling or transferring receivables to a related entity or by treating them as long-term receivables.	**Reduces days' sales outstanding** and enhances the receivables turnover ratio. May be done to mask aggressive revenue recognition practices.
Reclassifying inventory as other (long-term) assets.	**Increases inventory turnover** ratio. Current ratio will decrease.
Reclassifying non-core revenues as revenues from core continuing operations.	**Misleads about the sustainability of future revenues.**
Reclassifying expenses as non-operating.	Causes analysts to **treat recurring expenses as one-time** costs.
Treating investing cash flows (e.g., sale of long-term assets) as operating cash flows.	Operating cash flow is considered to be recurring and increasing it may lead to **higher equity valuation**.

Student's notes

[LOS 5.b] Explain potential problems that affect the quality of financial reports

Biased accounting example 1 – Misstated profitability

Mechanisms	Warning signs
• Aggressive revenue recognition, including channel stuffing bill and hold sales and outright fake sales. • Lessor use of finance lease classification. • Classifying non-operating revenue/income as operating, and operating expenses as non-operating. • Channeling gains through net income and losses through OCI.	▪ Revenue growth higher than peers'. ▪ Receivables growth higher than revenue growth. ▪ High rate of customer returns. ▪ High proportion of revenue is received in final quarter. ▪ Unexplained boost to operating margin. ▪ Operating cash flow lower than operating income. ▪ Inconsistency in operating versus non-operating classification over time. ▪ Aggressive accounting assumptions (e.g., high estimated useful lives). ▪ Executive compensation largely tied to financial results.

Student's notes

[LOS 5.b] Explain potential problems that affect the quality of financial reports

Biased accounting example 2 – Misstated assets/liabilities

Mechanisms	Warning signs
• Inappropriate models and/or inputs, affecting estimated values of financial statement elements (e.g., estimated useful lives for long-lived assets). • Reclassification from current to non-current. • Over- or understating allowances and reserves. • Understating identifiable assets (and overstating goodwill) in acquisition method accounting for business combinations.	▪ Inconsistency in model inputs for valuation of assets versus valuation of liabilities. ▪ Typical current assets (e.g., inventory, receivables) being classified as non-current. ▪ Allowances and reserves differ from those of peers and fluctuate over time. ▪ High goodwill relative to total assets. ▪ Use of special purpose entities. ▪ Large fluctuations in deferred tax assets/liabilities. ▪ Large off-balance-sheet liabilities.

Biased accounting example 3 – Misstated operating cash flow

Mechanisms	Warning signs
• Managing activities to affect cash flow from operations (e.g., stretching payables). • Misclassifying investing cash flow as cash flow from operations.	▪ Increase in payables combined with decreases in inventory and receivables. ▪ Capitalized expenditures (which flow through investing activities). ▪ Increases in bank overdraft.

Student's notes

[LOS 5.b] Explain potential problems that affect the quality of financial reports

Quality issues & Mergers and Acquisitions

Mergers and acquisitions

Opportunities for companies to manage financial results (through acquisition method)

- Acquiring companies may try to manipulate earnings upward after an acquisition if they want to positively influence investors' opinion of the acquisition.
- Target companies may be motivated increase reported earnings to secure a more favorable price for their company.
- Stock acquisitions provide an incentive for the acquiring company management to pursue aggressive accounting
- Underestimate the value of identifiable net assets & overestimating goodwill → decrease depreciation cost and improve earnings

GAAP accounting but not economic activity

Sometimes, an accounting treatment may conform to reporting standards but result in financial reporting that does not faithfully represent economic reality

→ **Example 1:** recognition of impairment or restructuring losses in a period reflects overstatement of income in prior periods

→ **Example 2**: avoid consolidation of various special purpose entities on technical grounds → keeping large losses and liabilities off-balance-sheet

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of a company's financial reports

Steps in evaluating the quality of financial reports

Understand the company, its industry, and the accounting principles and why such principles are appropriate.

↓

Understand management including their compensation; evaluate any insider trades and related party transactions.

↓

Identify material areas of accounting that are vulnerable to subjectivity.

↓

Cross-sectional and time series comparisons of FS and important ratios.

↓

Check for warning signs as discussed previously.

↓

For firms in multiple lines of business or for multinational firms, check for shifting of profits or revenues to a specific part of the business that the firm wants to highlight.

↓

Use **quantitative tools** to evaluate the likelihood of misreporting.

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of…

Beneish model overview (Quantitative tools)

Beneish Model

M-score = – 4.84 + 0.920 (DSRI) + 0.528 (GMI) + 0.404 (AQI) + 0.892 (SGI) + 0.115 (DEPI) − 0.172 (SGAI) + 4.679 (Accruals) − 0.327 (LEVI)

where:

M-score > –1.78 (i.e., less negative) indicates a higher-than-acceptable probability of earnings manipulation.

Intepretation

The M-score in the Beneish model is a normally distributed random variable with a mean of 0 and a standard deviation of 1.0. In order to determine the probability of earnings manipulation suggested by an M-score, one simply needs to look up the normal distribution statistical table. For example, an M-score of -1.49 indicates that the probability of earnings manipulation is 6.8%.

Limitation

The Beneish model **relies on accounting data**, which may **not reflect economic reality**. Deeper analysis of underlying relationships may be warranted to get a clearer picture. Additionally, as managers become aware of the use of specific quantitative tools, they may begin to game the measures used.

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of…

M-score – Beneish model

- **Days sales receivable index (DSRI)**: Days of sales receivables Year T vs. Year T – 1

  Large increase in DSRI could be indicative of revenue inflation.
- **Gross margin index (GMI):** Gross margin Year T – 1 vs. Year T.

  GMI > 1, the gross margin has deteriorated. A firm with declining margins is more likely to manipulate earnings.
- **Asset quality index (AQI):** (Non-current assets - PPE)/Total assets Year T vs. Year T – 1

  Increases in AQI could indicate excessive capitalization of expenses
- **Sales growth index (SGI):** Sales in Year T vs. Year T – 1

  Growth companies tend to find themselves under pressure to manipulate earnings to meet ongoing expectations.
- **Depreciation index (DEPI):** Depreciation rate Year T – 1 vs. Year T → The depreciation rate is depreciation expense /by (Depreciation + PPE)

  DEPI > 1 suggests that assets are being depreciated at a slower rate in order to manipulate earnings.
- **Sales, general and admin expenses index (SGAI):** SGA margin Year T vs. Year T – 1

  Increases in SGA margin might predispose companies to manipulate.
- **Accruals** = (Income before extraordinary items – CFO) / Total assets.
- **Leverage index (LEVI):** Total debt / Total assets Year T vs. Year T – 1

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of...

Quantitative tools – Beneish model example

Question:

Pritesh Deshmukh, CFA is analyzing the financial statements of Baza Restaurants Inc. Deshmukh wants to use the Beneish model to evaluate the probability of earnings manipulation. Deshmukh makes the following statements:

1. Depreciation index of less than 1 would indicate that the company is depreciating assets at a lower rate than in prior years.
2. Sales growth index of more than 1 indicates revenue inflation.

Which of the statements by Deshmukh are most accurate?

A) Statement 2 only

B) Statement 1 only

C) None of the statements is accurate

Depreciation index = $\frac{\text{Depreciation rate}_{t-1}}{\text{Depreciation rate}_{t}}$; Depreciation rate = Depreciation/(Depreciation + PPE)

Depreciation index < 1 => depreciation rate of current year > depreciation rate of prior years => Statement 1 is incorrect

Sales growth index = $\frac{\text{Sales}_{t}}{\text{Sale}_{t-1}}$ => SGI > 1 => positive growth, but this is not a sign of revenue inflation

$\Rightarrow$ **C is correct.**

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of…

Beneish model example (continued)

Variable	Value
DSRI	1.19
GMI	0.88
AQI	0.90
SGI	1.12
DEPI	1.19
SGAI	0.78
Accruals	0.12
LEVI	0.55
M-Score	-1.53
Probability	9.58%

Question:

Beneish's M-Score analysis for Pattern Processors Inc is shown in the following:

1. Using a cutoff value of –1.78 for the M-score, what would you conclude about the probability of earnings manipulation for PPI?
2. What are the implications of the DSRI and DEPI variables for PPI?

Answer

M-score > -1.78 => a higher-than-acceptable probability of earnings manipulation. The estimated probability of earnings manipulation is 9.58%.

Both DSRI & DEPI > 1 => the firm may be accelerating revenue recognition and accelerating revenue recognition

Student's notes

[LOS 5.c,d] Describe how to evaluate the quality of a company's financial reports; Evaluate the quality of…

Altman model - Quantitative tools

Formula

Z-score = **1.2** (Net working capital/Total assets) + **1.4** (Retained earnings/Total assets) + **3.3** (EBIT/Total assets) + **0.6** (Market value of equity/Book value of liabilities) + **1.0** (Sales/Total assets)

Interpretation

Higher is Better

1.81> Z-score	: high probability of bankruptcy
1.81< Z-score <3	: unclear indicators
3 < Z-score	: low probability of bankruptcy

Limitation

- It uses only one set of financial measures which are taken at a single point in time.
- Financial statements measure past performance, and reported balance sheet values assume that the company is a going concern rather than one that may fail.

[LOS 5.e,f] Describe indicators of earnings quality; Describe the concept of sustainable (persistent) earnings

Indicators of earnings

Indicators	Recurring earnings	
	Mean reversion in earnings	
	Earnings persistence & measures of accruals	$earnings_{t+1} = \alpha + \beta_1 . cashflow_t + \varepsilon$
		$earnings_{t+1} = \alpha + \beta_1 . cashflow_t + \beta_2 . accruals_t + \varepsilon$
	Beating benchmark	
	External factors	Eg: enforcement actions by regulatory authorities, restatements of previously issued financial statements

Student's notes

Student's notes

[LOS 5.g] Explain mean reversion in earnings and how the accruals component of earnings affects the speed of mean reversion

Indicators of earning-recurring earning, mean reversion and accruals

Recurring earning

High proportion of non-recurring items → Less sustainable low quality

Mean reversion in earnings

Extreme levels of earnings reverting over time → Mean reversion in earnings → One cannot simply extrapolate either very high or very low earnings into the future

Earnings with **high accruals** → **Higher speed** → Mean reversion in earnings

Student's notes

[LOS 5.g] Explain mean reversion in earnings and how the accruals component of earnings affects the speed of mean reversion

Accruals

Andre Bursh, is analyzing large retailers and has collected the following information on three companies based on the most recent financial statements

	Allied Stores	Beta Mart	Cash N Carry
Total earnings (per share)	$2.80	$1.33	$0.75
Cash element	$1.90	$0.78	$0.25
Accrual element	$0.90	$0.55	$0.50

Bursh notes that all three companies have reported stellar earnings this past year. Bursh is concerned about sustainability of such high earnings. Which company's earnings will revert to its mean fastest?

A) Beta Mart

B) Allied Stores

C) Cash N Carry

Answer:
Proportion of accruals in earnings: Allied Stores – 32%, Beta Mart – 31.8%, Cash N Carry – 66.7% => Cash N Carry's earnings will revert to normal fastest

[LOS 5.h] Evaluate the earnings quality of a company

Steps in the analysis of revenue recognition practices

Subjectivity in revenue recognition

↓

Vulnerability to **manipulation**

Understand the basics

↓

Evaluate and question ageing receivables

↓

Cash vs. Accruals

↓

Compare financials with physical data provided by the company

↓

Evaluate revenue trends and compare with peers.

↓

Check for related party transactions.

Student's notes

Student's notes

[LOS 5.h] Evaluate the earnings quality of a company

Expense capitalization issues

Under-report an operating expense by **capitalizing** it → Performance boosted → Analysts should **question** whether there is a systematic capitalization of expenses underway

Evaluating Procedures

Understand the basics

Cost capitalization policy, depreciation policy

↓

Trend and comparative (peer) analysis

↓

Check for related party transactions

[LOS 5.i] Evaluate the cash flow quality of a company

Indicators of cashflow quality

Cash flow quality (most importantly operating cash flow)

- Reporting quality
- Managers’ behaviors
- Economic performance
- Corporate life

High quality cash flow

Positive OCF from sustainable source and low volatility **(economic performance)**	Capital expenditures, dividends, and debt repayments covered **(managers' behavior)**	Small difference between earning and cashflow **(managers’ behaviors, reporting quality)**

Manager manipulation, cashflow shifting issues need to be considered carefully by practice such as reviewing activity ratio

Student’s notes

[LOS 5.j] Describe indicators of balance sheet quality

Evaluating Procedures

Checking for any unusual items or items that have not shown up in prior years.

Checking revenue quality

Example: Aggressive revenue recognition practices typically result in an increase in receivables—which reduces operating cash flow.

Checking for strategic provisioning

Example: Provisions for restructuring charges show up as an inflow (a non-cash expense) in the year of the provision and then as an outflow when ordinary operating expenses are channeled through such reserves.

Note: Analyst should also pay attention to the treatment of certain items (Example: dividends/interest paid can be identified as operating or investing cashflow under IFRS, firms also have significant flexibility in classifying securities)

Student's notes

Student's notes

[LOS 5.k,l] Evaluate the balance sheet quality of a company; Describe indicators of cash flow quality

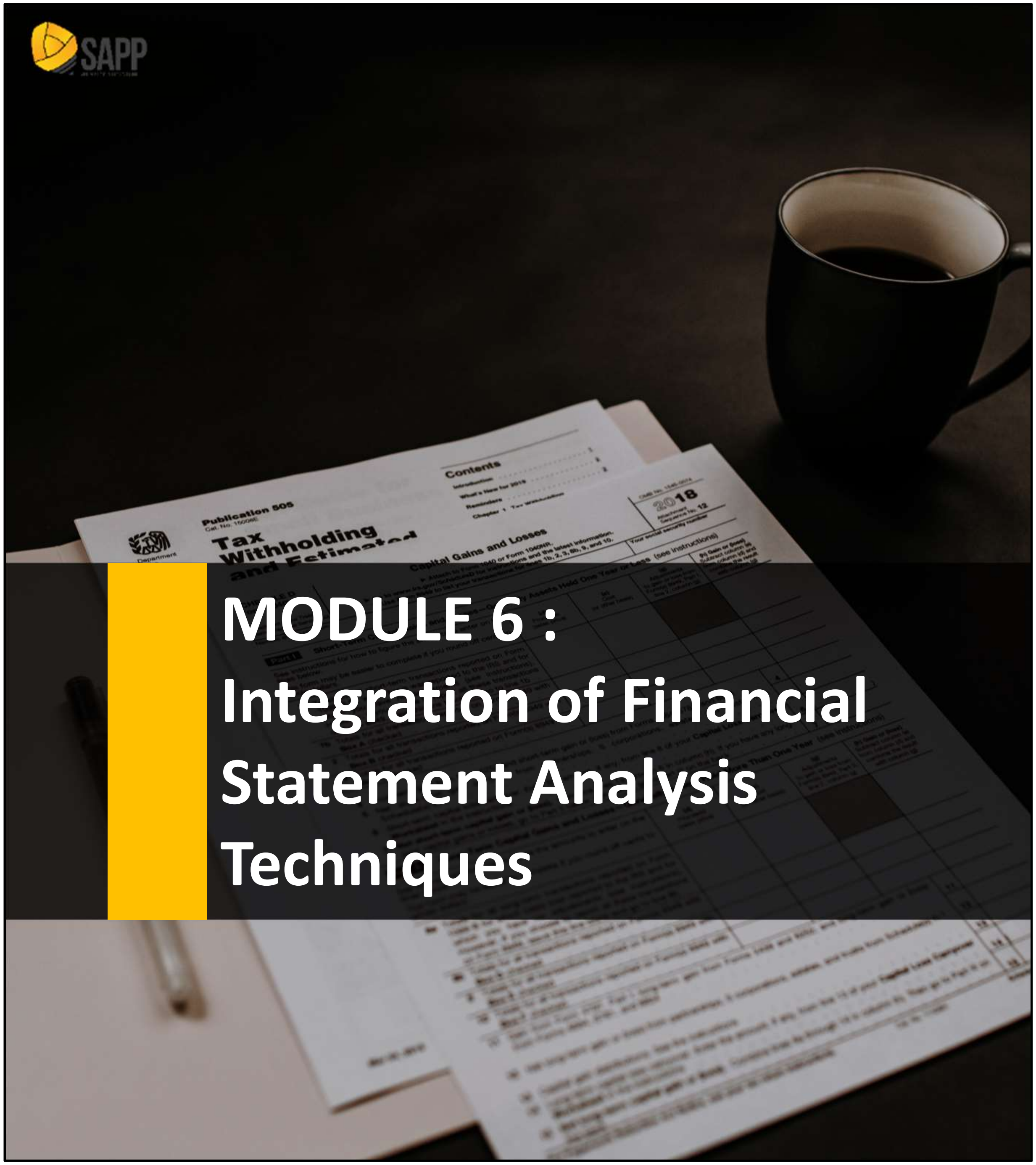
The balance sheet quality of a company

Completeness	Unbiased measurement	Clear presentation
• Off-balance-sheet obligations • The parent company liability in the investee • Higher profit margin because of investment in other companies	• Value of the pension liability • Value of investment in other companies • Goodwill value • Inventory valuation • Impairment of PPE and other assets	Clarity should be **evaluated in conjunction with information found in the notes** to financial statements and supplementary disclosures

[LOS 5.m] Describe sources of information about risk

Sources of information about risk

- Auditor's opinion
- Notes to the financial statements
- Management commentary
- Open required disclosures
- Financial Press



MODULE 6 : Integration of Financial Statement Analysis Techniques

Student's notes

Student's notes

LEARNING OUTCOMES

6.a. Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose (e.g., valuing equity based on comparables, critiquing a credit rating, obtaining a comprehensive picture of financial leverage, evaluating the perspectives given in management's discussion of financial results)

6.b. Identify financial reporting choices and biases that affect the quality and comparability of companies' financial statements and explain how such biases may affect financial decisions

6.c. Evaluate the quality of a company's financial data and recommend appropriate adjustments to improve quality and comparability with similar companies, including adjustments for differences in accounting standards, methods, and assumptions

6.d. Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios

6.e. Analyze and interpret how balance sheet modifications, earnings normalization, and cash flow statement related modifications affect a company's financial statements, financial ratios, and overall financial condition

Student's notes

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given a particular problem, question, or purpose

Step	Input	Output
Step 1: Establish the objectives	Perspective of the analyst Needs or concerns communicated by the client or supervisor Institutional guidelines	Purpose statement Specific questions to be answered Nature and content of the final report Timetable and resource budget
Step 2: Collect data	Financial statements Communication with management, suppliers, customer,….	Periodically updated information
Step 3: Process data	Data from Step 2	Adjusted financial statements Common-size statements Ratios Forecasts
Step 4: Analyze data	Data from Step 2 and 3	Results
Step 5: Develop and communicate conclusions	Results from analysis Published report guidelines	Report answering questions posed in Step 1 Recommendations
Step 6: Follow up	Periodically updated information	Nature and content of the final report

[LOS 6.a] Demonstrate the use of a framework for the analysis of financial statements, given…

Step 4: Analyze data

The focus of analyzing data

1) Sources of earning and ROE
2) Asset base
3) Capital structure
4) Capital allocation decision
5) Earnings quality and cash flow analysis
6) Market value decomposition
7) Anticipating changes in accounting standards

Student's notes

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and recommend appropriate adjustments

Example to illustrate the whole reading

Thunderbird is a manufacturing firm and own a 30% equity interest in one of its suppliers, Eagle Corporation

$ in millions	2016	2015	2014	2013
Income statement				
Revenue	$75,286	$68,921	$63,781	NA
EBIT	10,517	9,311	8,313	NA
EBT	9,463	8,474	7,258	NA
Income from associates and joint venture(*)	896	674	627	NA
Net income	7,967	6,894	6,023	NA
Balance sheet				
Total assets	$80,261	$71,264	$71,903	$61,731
Equity method investment	6,255	5,901	4,951	3,638
Stockholder equity	37,964	36,994	34,348	27,382

(*): not included in EBT, EBIT but included in net income

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

DuPont analysis

ROE

Tax Burden	Interest burden	EBIT margin
$\frac{NI}{EBT}$	$\frac{EBT}{EBIT}$	$\frac{EBT}{Revenue}$

Total Asset Turnover	Financial Leverage
$\frac{Revenue}{Average\ assets}$	$\frac{Average\ assets}{Average\ equity}$

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Extended DuPont Analysis - Thunderbird example

	Tax burden	Interest burden	EBIT Margin	Total asset turnover	Financial Leverage	ROE
2014	82.98%	87.31%	13.03%	0.955	2.165	19.51%
2015	81.35%	91.01%	13.51%	0.963	2.007	19.33%
2016	84.19%	89.98%	13.97%	0.994	2.021	21.26%

Extended DuPont Analysis (Excluding Equity Income and Investment Asset)

	Tax burden	Interest burden	EBIT Margin	Total asset turnover	Financial Leverage	ROE
2014	74.35%	87.31%	13.03%	1.020	2.165	18.68%
2015	73.40%	91.01%	13.51%	1.042	2.007	18.87%
2016	74.72%	89.98%	13.97%	1.080	2.021	20.51%

→ **Thunderbird's ROE is lower when excluding its associate !!**

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Analysis of the asset base requires an examination of changes in the composition of balance sheet assets over time. Presenting balance sheet items in **a common-size** format is a useful starting point.

$ in millions	2016		2015		2014	
Current assets	$25.039	31.2%	$24,714	34.7%	$29,236	40.7%
PP&E	15,445	19.2%	14,161	19.9%	13,293	18.5%
Identifiable intangibles	5,052	6.3%	2,641	3.7%	1,996	2.8%
Good will	23,396	29.1%	19,959	28.0%	18,893	26.3%
Other non current assets	11,329	14.1%	9,789	13.7%	8,485	11.8%
Total assets	$80,261		$71,264		$71,903	

Analysis: As a manufacturing firm Thunderbird is expected to have considerable investments in both current assets and fixed assets . However, note the significance of goodwill, which is 29.1% of total assets at the end of 2016 goodwill has increased since 2014, indicating Thunderbird has completed a number of business acquisitions goodwill is no longer amortized through the income statement, we must consider the possibility of losses in the future if goodwill is determined to have been impaired.

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

A firm's capital structure must be able to **support management's strategic objectives** as well as to allow the firm to honor its future obligations.

Thunderbird's long-term capital

$ in millions	2016		2015		2014	
Long term debt	4,290	8.6%	4,886	10.0%	5,794	12.4%
Other long term liabilities	7,679	15.4%	6,669	13.7%	6,693	14.2%
Stockholder equity	37,964	76.0%	36,994	76.2%	34,348	73.4%
Total long term capital	49,933		48,529		46,805	

Analysis: Financial leverage ratio does not reveal the true nature of the leverage, as some liabilities are more burdensome than others. Thunderbird's long-term debt has decreased from 12.4% of long-term capital in 2014 to 8.6% in 2016, a significant decrease in financial leverage. Given that Thunderbird's long-term debt has decreased, we need to consider the possibility of an offsetting change in the firm's working capital.

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Consolidated financial statements

↓

Individual characteristics of dissimilar subsidiaries can be **hided**

↓

Firms are required to disaggregate financial information

↓

Required disclosures about each segment's contribution to revenue and profit, the relationship between capital expenditures and rates of return, and which segments should be **de - emphasized or eliminated**

Student's notes

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Thunderbird EBIT and Revenue by segments

$ in millions	2016		2015		2014	
			Revenue			
Aircraft	$11,027	14.6%	$8,856	12.8%	$7,863	12.3%
Automotive	34,631	46.0%	32,754	47.5%	30,276	47.5%
Marine	22,345	29.7%	20,566	29.8%	19,941	30.6%
Specialty	7,283	9.7%	6,745	9.8%	6,151	9.6%
	$75,286		$68,921		$63,781	
			EBIT			
Aircraft	$2,440	23.2%	$1,995	21.0%	$1,696	20.4%
Automotive	5,059	48.1%	4,674	50.2%	4,115	49.5%
Marine	2,482	23.6%	2,160	23.2%	2,020	24.3%
Specialty	536	5.1%	522	5.6%	482	5.8%
	$10,517		$9,311		$8,313	

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Thunderbird EBIT and Revenue by segments

$ in millions	2016		2015		2014	
Asset(*)						
Aircraft	$14,777	27.54%	$6,861	15.40%	$5,288	12.85%
Automotive	20,059	37.39%	19,533	43.85%	19166	46.56%
Marine	12,310	22.94%	11,927	26.78%	10,779	26.19%
Specialty	6,509	12.13%	6,219	13.96%	5,928	14.40%
	$53,655		$44,540		$41,161	
Capial expenditures						
Aircraft	$383	11.31%	$336	11.80%	$240	10.45%
Automotive	1,432	42.29%	1,199	42.10%	1,018	44.32%
Marine	841	24.84%	667	23.42%	618	26.90%
Specialty	730	21.56%	646	22.68%	421	18.33%
	$3,386		$2,848		$2,297	

(*): Not equal to total assets due to unallocated and non-segment assets.

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

	EBIT margin			CapEX %/ Assets %		
	2016	2015	2014	2016	2015	2014
Aircraft	22.1%	22.1%	21.6%	0.41	0.77	0.81
Automotive	14.6%	14.3%	13.6%	1.13	0.96	0.95
Marine	11.1%	10.5%	10.4%	1.08	0.87	1.03
Specialty	7.4%	7.7%	7.8%	1.79	1.62	1.27

Depreciation and amortization by segment

$ in millions	2016	2015	2014
Aircraft	$172	$165	$142
Automotive	$644	$590	$603
Marine	$377	$328	$366
Specialty	$329	$318	$251
Total	$1,522	$1,401	$1,362

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and...

Thunderbird example

Estimated cash flow = EBIT + Depreciation/Amortization

$ in millions	2016	2015	2014
Aircraft	$2,612	$2,120	$1,838
Automotive	$5,703	$5,264	$4,713
Marine	$2,859	$2,488	$2,386
Specialty	$865	$840	$733
Total	$12,039	$10,712	$9,675

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Thunderbird example

EBIT and Revenue by segments analysis
The automotive division is the largest segment while the specialty products division is the smallest. Also, the percentage contribution to EBIT by the specialty products division declined from 5.8% in 2014 to 5.1% in 2016.

Assets and Capital Expenditures by Segment analysis

- The automotive division requires the greatest proportion of assets and capital expenditures.
- The specialty products division has the least assets of all four divisions and that the aircraft division has required the least capital expenditures.
- The specialty products division has the least assets of all four divisions and that the aircraft division has required the least capital expenditures.

Student's notes

[LOS 6.b,c] Identify financial reporting choices and biases; Evaluate the quality of a company's financial data and…

Thunderbird example

EBIT Margin and CapEx % to Assets % analysis

The specialty products division hasthe lowest EBIT margin but it has the highest capital expenditures proportion to assets proportion ratio => the specialty products division has, by far, the lowest EBIT margin, it has the highest capital expenditures proportion to assets proportion ratio.

Cash generation analysis

Capital is poorly allocated to the specialty division. The aircraft division—while continuing to produce superior operating margins—has fallen behind in cash generation in the latest year.

[LOS 6.d,e] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements and ratios; Analyze and interpret how balance sheet modifications, earnings normalization, and cash flow statement related modifications affect a company's financial statements, financial ratios, and overall financial condition

Earnings quality

Level of earnings	Earning sustainability

High quality earnings are close to cash flow
Earnings are subject to accrual accounting events that require numerous judgments and estimates.
⇒ earnings are subject to accrual accounting events that require numerous judgments and estimates.

Student's notes

Student's notes

[LOS 6.d,e] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements…

Accruals ratio analysis	Cashflow statement approach	$\text{Accruals}^{CF} = \text{NI-CFO-CFI}$
		$\text{Accruals ratio}^{CF} = \dfrac{\text{NI}-\text{CFO}-\text{CFI}}{(\text{NOA}_{END}+\text{NOA}_{BEG})/2}$
	Balance sheet approach	$\text{Accruals}^{BS} = \text{NOA}_{END}\text{-}\text{NOA}_{BEG}$
		$\text{Accruals ratio}^{BS} = \dfrac{(\text{NOA}_{END}-\text{NOA}_{BEG})}{(\text{NOA}_{END}+\text{NOA}_{BEG})/2}$

Note: In order to compare the two measures, it is necessary to eliminate cash paid for interest and taxes from operating cash flow by adding them back. This adjusted figure is the cash generated from operations (CGO).
CGO = EBIT + non-cash charges – increase in working capital.

NOA: Net operating assets = operating assets – operating liabilities
Operating assets = total assets – cash and cash equivalents
Operating liabilities = total liabilities – total debt

Student's notes

[LOS 6.d,e] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements…

Question

Suppose that we are provided the following financial data for MegaCo Industries:

Operating income	20,000
Net income	12,000
Cash from operations	33,000
Cash from investing	(30,000)
Cash from financing	(10,000)
Average total assets	90,000
Average net operating assets	180,000

MegaCo's cash flow accruals ratio is closest to:

A. 5.0%. B. 7.5%. C. 10.0%.

Answer

Cash flow accruals ratio = (NI – CFO – CFI) ÷ Average NOA = [12,000 – 33,000– (–30,000)] / 180,000 = 5%.

Student's notes

[LOS 6.d,e] Evaluate how a given change in accounting standards, methods, or assumptions affects financial statements…

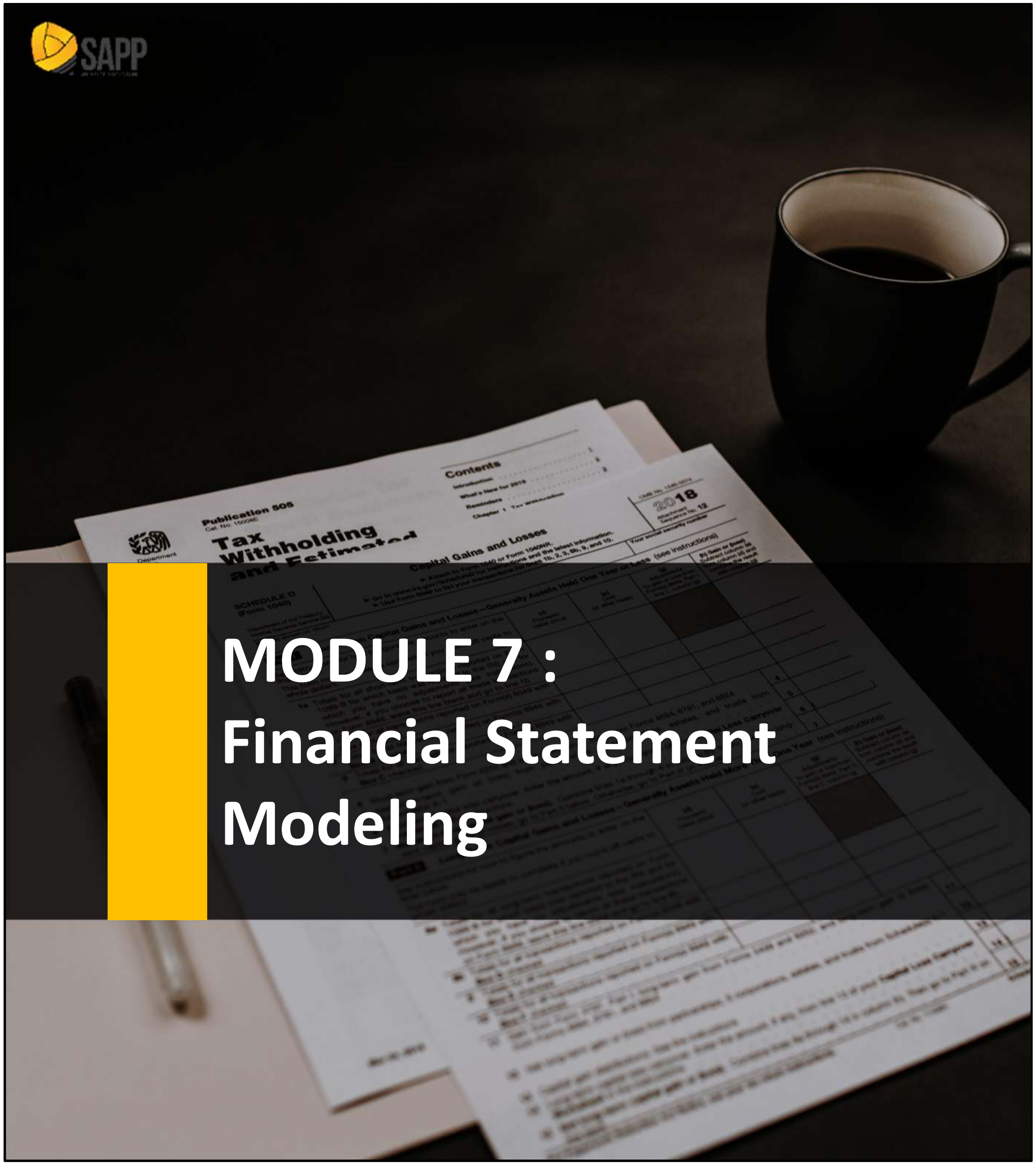
Market decomposition

Standalone value of the parent is the implied value of the parent without regard to the value of the associate.

Implied value = **Parent's market value** − **parent's pro-rata share of the associate's market value.**

As noted earlier, Thunderbird owns a 30% equity interest in Eagle Corporation, a publicly traded firm located in Europe. Let's suppose that the market capitalization of Thunderbird is \$137 billion. Also, let's imagine that the market capitalization of Eagle is €60 billion and, at year-end, the \$/€ exchange rate is \$1.40.

In this case, Thunderbird's pro-rata share of Eagle's market value is \$25.2 billion (€60 billion × 30% × \$1.40). Therefore, the implied value of Thunderbird, excluding Eagle, is \$111.8 billion (\$137 billion − \$25.2 billion) or 81.6% of Thunderbird's market capitalization (\$111.8 billion / \$137 billion).



MODULE 7 : Financial Statement Modeling

Student's notes

Student's notes

LEARNING OUTCOMES

7.a. Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models

7.b. Compare "growth relative to GDP growth" and "market growth and market share" approaches to forecasting revenue

7.c. Evaluate whether economies of scale are present in an industry by analyzing operating margins and sales levels

7.d. Demonstrate methods to forecast cost of goods sold and operating expenses

7.e. Demonstrate methods to forecast non-operating items, financing costs, and income taxes

7.f. Describe approaches to balance sheet modeling

7.g. Demonstrate the development of a sales-based pro forma company model

7.h. Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases

LEARNING OUTCOMES

7.i. Explain how competitive factors affect prices and costs

7.j. Evaluate the competitive position of a company based on a Porter's five forces analysis

7.k. Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation

7.l. Evaluate the effects of technological developments on demand, selling prices, costs, and margins

7.m. Explain considerations in the choice of an explicit forecast horizon

7.n. Explain an analyst's choices in developing projections beyond the short-term forecast horizon

Student's notes

Student's notes

[LOS 7.a,b] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models; Compare "growth relative to GDP growth" and "market growth and market share" approaches to forecasting revenue

Approaches to project future revenue

Top-down approach	**Hybrid approach**	**Bottom-up approach**
Begins at the level of the *overall economy*		Begins at the level of the *individual company*

Student's notes

[LOS 7.a,b] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models…

Top-down analysis

Forecast on overall economy, macroeconomic variable

Forecast on specific sector, market

Projections for individual company

Two common top-down approaches

Growth relative to GDP growth

Relationship between GDP growth and company sales:

1. Percentage point premium (or discount): *growth x% faster*

sales growth = GDP growth + x%

2. Relative terms: *growth at a x% faster rate*

sales growth = growth rate of GDP × (1 + x%)

Market growth and market share

Market growth = Industry sales

Market share = % of industry sales

Company revenue = Market growth × Market share

Student's notes

[LOS 7.a,b] Compare top-down, bottom-up, and hybrid approaches for developing inputs to equity valuation models…

Bottom-up analysis

Overall forecasts

Total company revenue forecast

Individual company/Unit within the company

Bottom-up approaches

- **Time series:** forecasts based on historical growth rates or time-series analysis.
- **Returns-based measure:** forecasts based on balance sheet accounts.
- **Capacity-based measure:** forecasts based on same-store sales growth (opened for at least 12 months) or sales related to new stores

Top-down analysis + Bottom-up analysis = **Hybrid approaches** → *Most common approaches*

Student's notes

[LOS 7.c] Evaluate whether economies of scale are present in an industry by analyzing operating margins and sales levels

Volume ↑ + Average cost of production ↓

Economies of scale

Firm's characteristics
- Higher operating margins
- Sales volume and margins tend to be positively correlated

Contributing factors
- High fixed costs
- Higher levels of production
- Greater bargaining power with suppliers
- Lower per unit advertising expense

<u>Evaluating the presence of economies of scale:</u> Analyzing common-size income statements

- Lower COGS/sales → Economies of scale in COGS
- Lower SG&A/Sales → Economies of scale in SG&A

Student's notes

[LOS 7.d] Demonstrate methods to forecast cost of goods sold and operating expenses

1. Forecasting cost of goods sold

Because COGS is closely related to revenue, future COGS is usually estimated as a *percentage of future revenue*:

Forecast COGS = (historical COGS / revenue) × (estimate of future revenue)

or

Forecast COGS = (1 – gross margin) × (estimate of future revenue)

Gross margin:

- Increasing or decreasing trend → Consider the probability the continuity of this trend.
- Firm's competitors' gross margins may provide a check of reasonableness of estimates.

Volume and price inputs: Closer examination may improve the quality of forecasts (e.g., firms hedge their future input costs by using derivative securities)

Separate forecasting: improve firm's COGS estimation by forecasting COGS for various product categories and business segments.

Student's notes

[LOS 7.d] Demonstrate methods to forecast cost of goods sold and operating expenses

2. Forecasting Selling General and Administrative Costs

Characteristics of SG&A

- SG&A operating expenses are *less sensitive* to changes in sales volume, fixed cost component is greater than its variable cost component (e.g., R&D expenditures, management salaries and IT operations, etc.)
- Selling and distribution costs may be more directly related to sales volumes

- Break out the *components* of SG&A and consider separately to improve the overall forecast of SG&A expenses
- *Difference segments* have significant differences in SG&A as a percentage of revenue → Forecast separately to product better estimate

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

1. Forecasting Financing costs

Level of gross debt + Market interest rates

Gross interest expense

Net debt = Gross debt – (cash + cash equivalent + short-term securities)

Net interest expense = Gross interest expense – interest income

Note: Analysts should also use any *planned debt issuance or retirement* and the *maturity structure of existing debt* (disclosed in footnotes to financial statements) to improve the forecasts of future financing costs

Student's notes

Student's notes

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

2. Forecasting Income Tax Expense

Primary tax rates

Statutory rate	**Effective tax rate**	**Cash tax rate**
The percentage tax charged in the country where the firm is domiciled	Income tax expense as a percentage of pretax income on the income statement	Cash taxes paid as a percentage of pretax income

Differences arise from:

- *Tax credits*
- *Withholding tax on dividends*
- *Adjustments to previous years*
- *Non-deductible expenses*

Analyst should pay attention to companies that consistently report *effective tax rate < statutory tax rate*

The difference

↓

Changes in Deferred tax items

Income tax expense = Cash tax due + ΔDTL = ΔDTA

Student's notes

[LOS 7.e] Demonstrate methods to forecast non-operating items, financing costs, and income taxes

3. Forecasting Other items

- **Dividends:**
  - Can be estimated based on historical data, using a constant growth rate or a constant payout ratio.
- **Number of outstanding shares**
  - Changes due to: dilution related to stock options, convertible bonds and similar securities, issuance of new shares, share repurchases,...
  - Projections for share issuance and repurchases should fit within the analyst's broader analysis of a company's capital structure.
- **Unusual charges**
  - Almost impossible to predict → Typically excluded from forecasts.
  - If certain recurring costs are classified as *unusual* → Should consider some normalized level of charges in the model.

[LOS 7.f] Describe approaches to balance sheet modeling

Income statement modeling

Balance sheet modeling

Cash flow modeling

Many balance sheet items flow from the forecasted income statement

- Analyst can focus on either the balance sheet or cash flow modeling.
- The third financial statement will naturally result from the construction of the other two.

Student's notes

Student's notes

[LOS 7.f] Describe approaches to balance sheet modeling

1. Working Capital accounts

- Working capital items such as accounts receivable, accounts payable, and inventory are very closely linked to income statement projections.
- They can be modelled with historical efficiency ratios or the relationship with income statement items, for example:

Inventory turnover:

$$\text{Inventory turnover} = \frac{\text{Forecasted COGS}}{\text{Inventory}}$$

→ Can be used to forecast the inventory value for the balance sheet.

Days sales outstanding: similarly, projected accounts receivable can be forecasted.

$$\text{Projected A/R} = \text{Days sales outstanding} \times \frac{\text{Forecasted sales}}{365}$$

Using this method, the relationship of working capital items and income statement items will be *preserved* and working capital items will *increase at the same rates* as revenues.

[LOS 7.f] Describe approaches to balance sheet modeling

2. Long-term assets

- Projections for PP&E and intangible assets are less directly tied to the income statement.
- PPE is determined by depreciation and capital expenditure (capex).

Cash flow statement

- Depreciation Amortization → *Historical depreciation, management's disclosures, levels of long-term assets*
- Capital expenditure → *Maintenance capital expenditures*; *Growth capital expenditures*

→ **Balance sheet**

PP&E

Intangible assets

Note: Maintenance capital expenditure forecasts should normally be higher than depreciation because of inflation

3. Future capital structure

Analysts must make assumptions about a company's future capital structure. Leverage ratios can be useful for projecting future debt and equity levels.

- Once projected income statements and balance sheets have been constructed, future cash flow statements can be projected.
- The analyst then perform sensitivity analysis for individual assumptions (scenario analysis) to examine the sensitivity of net income to changes in assumptions.

Student's notes

[LOS 7.h] Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases

	Definition	Consequence	Mitigation
Overconfidence bias	Having *too much faith* in one's own work	May underestimate forecasting errors → narrower confidence interval of forecasts	Share forecasts, critiques, evaluate past forecasts, scenario analysis
Illusion of control bias	A false sense of security in one's forecasts	Overfitted models → perform poorly out of sample, conceal outdated assumption	Focus on variables with known explanatory power, seeking outside opinions
Conservatism bias	Making *small adjustments* to prior forecasts when new info is available	Reluctance to incorporate negative news, lags in incorporating positive news	Periodic evaluation of forecasting errors, flexible models
Representative -ness bias	Classifying data based on past info and known classifications	Incorrect understanding that biases all future thinking on the info	Start with base rate and determining differentiating factors
Confirmation bias	Pay attention to data that affirms their earlier beliefs	Pursue only positive news or certain criteria, narrow scope	Read and seek researches with contradicting views and opinions

Student's notes

[LOS 7.i,j] Explain how competitive factors affect prices and costs; Evaluate the competitive position of a company based on a Porter's five forces analysis

- Balance sheet
- Income statement
- Cash flow statement

Financial projections

Return on invested capital (ROIC)

$$= \frac{\text{NOPLAT}}{\text{Invested capital}}$$

NOPLAT: net operating profit adjusted for taxes

Invested capital = operating assets − operating liabilities

Firms with higher ROIC are likely to exploiting some **competitive advantage**

Porter's Five Forces

- **01 Threat of substitutes**
- **02 Rivalry**
- **03 Bargaining power of suppliers**
- **04 Barganing power of customers**
- **05 Threat of new entrants**

Student's notes

[LOS 7.k] Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation

Changes in input costs → Changes in earnings

1. Hedge with derivatives

Companies with commodity-type inputs can hedge their exposure to changes in input prices through derivatives → Reduce the effect of short-term changes.

Note: Vertically integrated companies are less subject to the effects of variations in input prices

2. Pass on to customer

Increase in input costs → Increase in product price → Sales volume and revenue may fall

The effects of increasing a product's price depend on the product's elasticity of demand *(Elastic demand: price increase → decrease in total sales revenue).*

If price *increase* but sales *do not decrease* → operating profit is unchanged, but gross margins, operating margins, net margins fall.

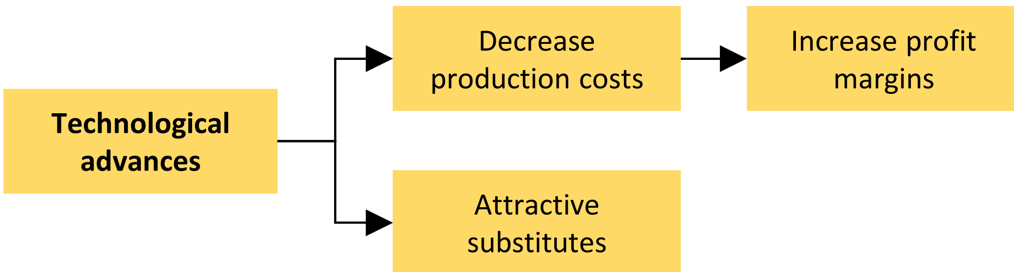
3. Substitute input

A firm can reduce the impact of an increase in an input price by switching to a substitute input *(e.g., rising oil prices may lead power generation firms to switch from oil to natural gas).*

Different businesses and geographic segments should be considered separately when appropriate, scenario and/or sensitivity analysis should be conducted.

Student's notes

[LOS 7.l] Evaluate the effects of technological developments on demand, selling prices, costs, and margins



Cannibalization factor: the percentage of new product sales that will replace existing product sales

$$\text{cannibalization rate} = \frac{\text{new product sales that replace the existing product sales}}{\text{total new product sales}}$$

Example:

Alpha, Inc., manufactures LED lightbulbs for sale to businesses as well as to households. Seventy-five percent of Alpha's unit sales are to business customers. Technological advances have enabled bulb manufacturers to produce a new bulb that is more energy efficient, and Alpha is planning to introduce a bulb next year that uses this new technology. Projected sales for the new bulb are shown in the following table.

Calculate the lost unit sales of current product due to cannibalization, assuming a cannibalization factor of 50% for businesses and 20% for households.

Market	Units
Businesses	87,000
Households	11,000
Total units	98,000

Cannibalization in business market segment = 87,000 × 0.50 = 43,000 units.

Cannibalization in household market segment = 11,000 × 0.20 = 2,200 units.

Total lost unit sales = 45,700 units

[LOS 7.m] Explain considerations in the choice of an explicit forecast horizon

Buy-side analyst

The appropriate forecast horizon may simply be the *expected holding period* for a stock.

E.g.: a portfolio with a 25% annual turnover, the average holding period of a stock is 4 years, so 4 years may be the most appropriate forecast horizon.

Highly cyclical companies

The horizon should be long enough that:

- The effects of the current phase of the economic cycle are not driving above-trend or below-trend earnings effects
- Include the middle of a business cycle so that the analyst's forecast includes mid-cycle level of sales and profits

Normalized earnings: expected mid-cycle earnings or expected earnings when the current effects of events or cyclicality are no longer affecting earnings

Impactful events

Events such as acquisitions, mergers, or restructurings, should be considered temporary.

The forecast horizon should be long enough that the perceived benefits of such events can be realized or not.

Student's notes

Student's notes

[LOS 7.n] Explain an analyst's choices in developing projections beyond the short-term forecast horizon

Valuing stock

Earnings /measure of cash flow + Stock's terminal value

Relative valuation / Price multiple

The multiple used must be consistent with the estimate of the company's growth rate and required rate of return.

E.g.: The average P/E ratio over the last 10 years assumes that earnings growth and required return will remain the same in the future.

Discounted cash flow

- **Expected earnings/CF** should be normalized to a mid-cycle value that is not affected by temporary initiatives and events
- **Expected future growth rate:** used to estimate terminal value as the present value of a perpetuity
- **Inflection point:** when the future will differ from the past

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Steps in developing a sales-based pro forma model

1	**Estimate revenue growth and future expected revenue**	Using market growth plus market share, trend growth rate, or growth relative to GDP growth.
2	**Estimate COGS**	Based on a percentage of sales, or on a more detailed method based on business strategy or competitive environment.
3	**Estimate financing costs**	Using interest rates, debt levels, and the effects of any large anticipated increases or decreases in CAPEX or anticipated changes in financial structure.
4	**Estimate income tax expense and cash taxes**	Using historical effective rates and trends, segment information for different tax jurisdictions, and anticipated growth in high- and low-tax segments.
5	**Estimate cash taxes**	Take into account changes in deferred tax items.

Student's notes

Student's notes

[LOS 7.g] Demonstrate the development of a sales-based pro forma company model

Steps in developing a sales-based pro forma model

7	**Model the balance sheet**	Based on items that flow from the income statement (working capital accounts – accounts receivable, accounts payable and inventory)
8	**Estimate CAPEX and net PP&E**	Using depreciation and CAPEX (for maintenance and for growth)
9	**Construct a pro forma cash flow statement**	Using completed pro forma income statement and balance sheet